

VIOLENCE AGAINST WOMEN AT WORK*

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We link every police report in Finland to administrative data to identify violence between colleagues and the economic consequences for victims, perpetrators, and firms. This new approach to observe when one colleague attacks another overcomes previous data constraints limiting evidence on this phenomenon to self-reported surveys that do not identify perpetrators. We document large, persistent labor market effects of between-colleague violence on victims and perpetrators. Male perpetrators experience substantially weaker consequences after attacking female colleagues. Perpetrators' relative economic power in male-female violence partly explains this asymmetry. Turning to broader implications for firm recruitment and retention, we find that male-female violence causes a decline in the proportion of women at the firm, both because fewer new women are hired and current female employees leave. Management plays a key role in mediating the effects on the wider workforce. Only male-managed firms lose women. Female-managed firms exhibit a key difference relative to male-managed firms: male perpetrators are less likely to remain employed after attacking their female colleagues. *JEL codes:* J16, J81, M54.

I. INTRODUCTION

The #MeToo movement demonstrated that violence between colleagues is an internationally relevant phenomenon. Yet there is little evidence on the effects of workplace-related violence on perpetrators, victims, and the wider workforce, nor how these

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effects may depend on power and gender differences between victims and perpetrators. The nascent economics literature on workplace sexual harassment has largely focused on less serious crimes, with a survey experiment featuring hypothetical cases of harassment suggesting potentially large costs to victims (Folke and Rickne 2022). However, due to data limitations, this literature has thus far been unable to identify the consequences that perpetrators of violence face for assaulting a colleague, nor the effects on the broader firm and the role of management.

We harness unique Finnish administrative data to analyze the effect of violence between colleagues on victims, perpetrators, and the broader firm. We link information on every police report in Finland between 2006 and 2019 to administrative records on employment, income, and demographic characteristics.¹ Crucially, our data contain unique identifiers for both the victim and perpetrator. This allows us to identify violent incidents where both parties worked in the same firm at the time of an incident. While police reports will miss the full scope of violent incidents among colleagues as it is likely that most cases of (minor) violence go unreported, they provide an important step in understanding this phenomenon.

This new methodological approach to observe when one colleague attacks another and what happens subsequently underpins the analysis in this article. It allows us to provide the first estimates of the economic consequences of realized events of violence between colleagues on victims and perpetrators, to understand how the economic relationship between the two in the firm affects outcomes, and to explore the broader effects of these events on firms, including the role of management. This innovation overcomes data constraints that restricted prior evidence on the topic from economics, psychology, and sociology to self-reports from surveys, which do not allow perpetrators to be identified.

We identify more than 4,600 cases of violence between colleagues. If we take seriously survey evidence that only 10% of assaults in Finland are reported to police, then 3% of individuals in the labor force are directly involved in colleague violence. Even more are affected through the spillover effects that we will document on the broader workforce. Fifty-five percent of incidents are assaults or petty assaults, and the remaining 45% are a mix

1. A police report initiates an investigation before a suspect is formally charged with a crime or a court case.

of negligent bodily injury, menace, and other crimes. The vast majority (84%) of perpetrators are men, and victims are evenly split between male and female. Victims of male-female violence are relatively low-earning women in the firm, compared with their perpetrators, who are relatively high-earning men. This is not the case for male-male between-colleague crimes, where violence is more likely to occur between relative equals in the firm. Compared to all firms in Finland, firms where violent incidents occur are larger and pay higher average wages. Firms that experience male-female violence have a similar share of female employees before the incident and are approximately as likely to be female-managed compared to all firms in Finland.

We first analyze how colleague violence affects victims' and perpetrators' outcomes. We begin with descriptive results, analyzing raw labor market outcomes in the years before and after the incident. A number of patterns immediately emerge. First, the earnings growth of victims and perpetrators is robust and indistinguishable from all other workers in Finland in the years leading up to the event. This is not consistent with years of unreported abuse causing a deterioration in labor market outcomes long before the reported event we observe. Second, we observe a sharp drop in employment for victims and perpetrators directly after an incident. There is an important asymmetry between male-female and male-male violence, with larger employment losses for female victims (compared with their male perpetrators) and smaller employment losses for male victims (compared with their male perpetrators).²

The precipitous employment losses following violence between colleagues in the raw data might not be caused by the violence itself. Some workers will always separate from their firm each year, and poor labor market potential might cause an individual to commit violence against a colleague or make a colleague more exposed to abuse. Such individuals may have experienced declines in employment even if they hadn't attacked (or been victimized by) a colleague. To mitigate these concerns, we employ a matched difference-in-differences design with individual fixed

2. We restrict our analysis of female-perpetrated workplace violence to [Online Appendix](#) Section B since (i) women rarely attack colleagues, resulting in small sample sizes, and (ii) in the majority of cases where a woman is recorded as the perpetrator, she is also recorded as a victim, thus these are not clear-cut cases of female-perpetrated violence.

effects. Specifically, we compare the evolution of employment outcomes of affected workers before and after an incident of workplace violence to observationally identical workers who were not affected.

This empirical strategy confirms the conclusions from our descriptive analysis. Following a violent incident, victims experience an immediate drop in employment that persists for at least five years. Perpetrators experience similarly large and persistent negative effects. The dramatic asymmetry in the impact of colleague violence for male-female crimes versus male-male crimes remains. For male-male crimes, results are as one might expect: perpetrators experience significantly greater negative repercussions than their victims. Employment rates fall by 12.3 percentage points for perpetrators and 7.5 percentage points for victims in the five years after a violent incident between men. Results are very different for male-female crimes. While perpetrator employment falls by only 7.1 percentage points on average in the five years after an incident, victim employment falls by 9.1 percentage points. This asymmetry is also apparent when we compare the effects of between-colleague violence to noncolleague violence: perpetrators who attack female colleagues are significantly more likely to keep their jobs compared with observationally identical perpetrators who attack women who are not colleagues.

We examine power differentials between victim and perpetrator as one possible explanation for this asymmetry in the effects of male-female versus male-male workplace violence. We interact the treatment of an attack by a colleague with an indicator for whether the perpetrator is a manager in the firm. We find that for male-female (male-male) crime, victims' employment rates fall by 7 (6.3) percentage points more when their perpetrator is a manager. However, perpetrators who are managers are 6 (13.7) percentage points less likely to be unemployed in the five years after an incident. Results are similar when using the difference in the within-firm income rank of perpetrators and victims as the measure of relative power, and when comparing to non-colleague violence. Thus, perpetrator power in the firm plays an important role in determining the effects of violence between colleagues, and we show that this partially accounts for the comparatively smaller labor market effects experienced by perpetrators of male-female violence where such power imbalances are more common. These results suggest that policies to support women moving up the occupational hierarchy may help mitigate these asymmetric impacts.

We investigate the broader implications of colleague violence for the firm. We find that male-female violence has systematic effects on the gender composition of the workplaces in which it occurs. After an incident, the gender composition of firms becomes significantly more male.³ This fall in the female share of employees only occurs in the case of male-female violence; there is no significant effect of male-male violence on the gender composition of employees. Moreover, there is no effect of noncolleague violence on the gender composition of the firm. The reduction in the share of employees who are women in firms where male-female violence takes place is explained by higher separation rates of existing female employees and a significant reduction in the share of women among new hires.

In the final part of the article, we focus on the role of managers in mediating the effects on the wider workforce. Previous research demonstrates that managers help determine the success of a firm (Bertrand and Schoar 2003; Bloom and Van Reenen 2007; Bloom et al. 2013, 2019; Bender et al. 2018; 2019; Bandiera et al. 2020). Moreover, there is important heterogeneity in how male and female decisionmakers interpret and respond to negative (or positive) shocks, and how the gender of the individual responsible for the shock might change the response of the manager (Sarsons 2017; Chakraborty and Serra 2021). In particular, Egan, Matvos, and Seru (2022) show that women found guilty of misconduct are more likely to be fired, and that this is entirely explained by the behavior of male-managed firms.

Motivated by these facts, we consider heterogeneity in the effects of colleague violence by the proportion of women in decision-making positions. Following Bender et al. (2018), we calculate the proportion of women in the top 20% of earners in the firm. We find that the reduction of women in the workforce is isolated to male-managed firms, that is, those that have a below-median share of women in high-earning positions. Perpetrators of workplace violence in female-managed firms are more likely to enter unemployment, although the gender composition of management does not significantly affect the direct victims' labor market outcomes. We find that it is both perpetrators losing their jobs, and female management more generally, that mitigates the consequences of colleague violence on the wider workforce.

3. This is true even when excluding victims and perpetrators from the analysis.

Our findings contribute to two broad strands of literature. Most closely related is a small but growing body of work showing that women disproportionately experience costly interactions with peers at work (Bertrand and Hallock 2001; Basu 2003; Antecol and Cobb-Clark 2006; Hersch 2011; Sarsons 2017). In particular, we complement Folke and Rickne (2022) who use survey data from Sweden to show that sexual harassment on the job is more likely for those in the gender minority, harassment is associated with higher turnover of victims, and that people presented with randomized hypothetical job offer vignettes in a survey setting are willing to give up approximately 10% of hypothetical wages to avoid workplace sexual harassment. This revealed preference to avoid harassment at work is consistent with our finding that there are very large costs to victimization, although their paper focuses on less severe forms of workplace harassment.

We complement their article by estimating the effects of realized events of between-colleague violence for both male and female victims and their perpetrators, as well as the effects on the broader firm. Given our unique data, we can link perpetrators and victims and examine how their relative economic standing mediates these effects. Our results suggest that the patterns characterizing high-profile male-female assaults reported in recent years to the media hold true in general; namely, there are large effects for victims and a relative lack of consequences for their male perpetrators, and this can be explained in part by perpetrators' relative power in the firm.

We also show that these events have effects that extend well beyond the perpetrator and the victim to the broader firm and that management gender composition can play a role in mitigating these broader costs. These broader repercussions for peers in the firm complement a large literature documenting spillovers among peers in the workplace (Thornton and Thompson 2001; Mas and Moretti 2009; Cornelissen, Dustmann, and Schönberg 2017; Nix 2020; Brune, Chyn, and Kerwin 2022), as well as in other contexts (Hoxby 2000; Bayer, Hjalmarsson, and Pozen 2009; Black, Devereux, and Salvanes 2013).

Second, this study makes an important contribution to the literature on firms and firm management on worker outcomes (Ichniowski, Shaw, and Prennushi 1995; Bertrand and Schoar 2003; Bandiera, Barankay, and Rasul 2007; Gosnell, List, and Metcalfe 2020; Alan, Corekcioglu, and Sutter 2023). We show that extreme and toxic events between colleagues result in large costs

for direct victims, especially female victims; cause other women in the firm to leave; and change the hiring patterns of firms. The gendered aspects of the firm's managerial response to these events is most closely related to [Sarsons \(2017\)](#), [Chakraborty and Serra \(2021\)](#), and [Egan, Matvos, and Seru \(2022\)](#) described previously, along with [Benson, Li, and Shue \(2021\)](#), who show that male managers are less likely to promote female subordinates who are equally productive by scoring them lower in terms of potential, and [Cullen and Perez-Truglia \(2019\)](#), who find that male managers are more likely to promote male subordinates.⁴

The article is organized as follows. [Section II](#) describes the data, how we measure colleague violence, and provides descriptive statistics on the characteristics of victims, perpetrators, and violent firms. [Section III](#) presents effects on victims and perpetrators; [Section IV](#) presents effects on firms. [Section V](#) explores the role of managers. [Section VI](#) concludes.

II. DATA AND DESCRIPTIVE STATISTICS

II.A. Data, Sample Construction, and Defining Workplace Violence

We use Finnish administrative data to study the effect of between-colleague violence on worker and firm outcomes. We obtained the universe of police reports filed between 2006 and 2019. A police report is the first step in any investigation and occurs before a perpetrator is formally charged with a crime and before any court case takes place. Reports can be filed online or in person at a police station. After an investigation, a suspect is charged only if the prosecutor considers that there is sufficient evidence to secure a conviction. Only after this step can a court case take place. Court cases are public records, but police reports are not.

The police reports include a unique case number for each reported crime; information on the day, month, year, and type of the offense; characteristics of the suspect (age, sex); characteristics of the victim (age, sex); and information on whether the case has been submitted to a prosecutor and whether the case has been solved. Note that there can be multiple charges associated with a given case (e.g., assault and menace). We focus on the total

4. Also related is [Stoddard, Karpowitz, and Preece \(2020\)](#), who show that women who find themselves in the minority in a group in a professional setting are at a disadvantage and are viewed less favorably by their peers.

number of cases, not the total number of charges. The police also indicate the most serious charge for a given case, and this is the crime type we use when constructing descriptive statistics.

Crucially, the police data includes unique national identification numbers for both the perpetrator and the victim when their identities are known. We use these numbers to link perpetrators and victims from the police data to the Finnish Linked Employer-Employee Data (FLEED) and FOLK data. FLEED is a population register data containing annual income, employment, and demographic characteristics. It covers every person alive in Finland aged 15–70 for 1988–2016. FOLK is identical but covers 2017–2019. This allows us to link everyone in the police data of working age to their demographic and employment information, including those who are outside the labor force, unemployed, or self-employed. Note that all links between the police data and FLEED/FOLK are perfect (i.e., there is no way to get “incorrect” links in contrast to other contexts where fuzzier matching using names and other information is used and matches may not be perfect). FLEED/FOLK also includes unique plant and firm identifying numbers for everyone working in Finland. Thus, with these linked data, we are able to identify police reports between colleagues employed at the same establishment, a key innovation of this study.⁵ Hereafter, we refer to the “establishment/plant” as the “firm,” as we do not focus on the broader firm identifiers (for example, we focus on each individual McDonald’s restaurant as the “firm” of interest as opposed to McDonald’s as a whole).

We construct the analysis sample in the following way. First, we merge police records from 2006–2019 to FLEED/FOLK data using victim and perpetrator unique identifiers and identify between-colleague violent incidents. Formally, we classify an incident recorded in a police report as between-colleague violence if two conditions are met. First, and most important, both the victim and perpetrator must have worked at the same firm in December just preceding the incident (we only observe the place of employment in December of each year). This requires that both the victim’s and the perpetrator’s identities are known such that their unique identifiers are recorded in the police data.⁶ Cases

5. Person, workplace, and firm identifiers are unique, ensuring matches are perfect.

6. Among all police reports, 75% include the identification of the perpetrators, that is, the personal IDs of perpetrators.

involving an unknown perpetrator (or unknown victim) will not be included. Second, we restrict our attention to cases with crime codes that capture any violent incident, including minor ones such as petty assaults and sexual harassment. We list all of the crime codes used to identify between-colleague violence in [Online Appendix A.4](#).

We do not require that a case be sent on to courts for inclusion in our sample, nor can we verify the veracity of the reports, for two reasons. First, there is currently no link between the police data and court outcomes.⁷ Second, evidence from criminology and sociology highlights significant case attrition between reporting and charging for gender-based violence that has little connection to the veracity of the case ([Hester 2006](#); [Spohn and Tellis 2012](#)).⁸ In a minority of cases, there are multiple perpetrators or multiple victims. We treat these as a single case for the purpose of counting the number of violent workplace incidents, include each perpetrator and victim separately for the individual analysis, and include the case only once in the firm analysis.

After merging the police data with our rich administrative data on individual labor market outcomes and creating the sample of between-colleague crimes, we construct the labor market trajectories of victims and perpetrators before and after an incident. For victims and perpetrators, we focus on two main outcomes: employment and income. Employment measures whether, and in which firm, an individual was working in the December of each calendar year. Income consists of all labor income reported to the tax authorities and measured in December each year (salary and wage earnings, as well as any self-employment income, but excluding benefits). When estimating effects on income, we estimate the effect of income measured in year t as a fraction of the average total labor market income in the five years before the event to address zeros. We use the plant identifiers from the full population register data to construct relevant firm outcomes such as headcount, turnover, and the gender composition of the

7. The link between the police data and court data is still ongoing. This linking is more complicated due to inconsistent inclusion of victim identifiers and lack of the exact date of the crime.

8. In addition, see anecdotal accounts from the #MeToo movement. For example, there was a 2015 police report for groping against Harvey Weinstein filed in 2015 by Ambra Gutierrez. She cooperated with police and wore a wire to gather evidence that confirmed her experience, yet the district attorney chose not to prosecute the case.

workforce. The plant identifiers allow us further to construct labor market outcomes of other workers employed at the same firm.

1. *Measurement Discussion.* The limited previous literature on between-colleague violence in other disciplines, such as sociology and psychology, almost solely relies on survey data and small selected samples. For example, an applied psychology literature suggests that violence and harassment are key workplace environment factors with implications for individual and organizational performance (Cortina et al. 2001; Estes and Wang 2008; Geck et al. 2017). Using administrative data to study this question not only allows us to examine direct effects on victims but makes it possible to examine outcomes that have never been studied in this context before, such as effects on perpetrators (prior papers have focused almost exclusively on victims), the role of the within-firm economic relationship between victim and perpetrator, effects on the broader firm including workplace colleagues and new recruits, and the role of management.

Our measure of between-colleague violence captures the population of violent incidents that occur between colleagues and are reported to the police. These incidents do not necessarily occur in the workplace itself: the police data do not include the precise location of the crime. This means that the incidents we study could be happening both inside and outside the official premises of the firm. This is an important advantage of our data relative to an alternative scenario where one only observes violence occurring during working hours within the four walls of the firm. People can be assaulted by colleagues at off-premises holiday parties, when traveling for work, and so on. Using our definition of between-colleague violence, all such incidents will be included. This means, however, that our measure will also capture domestic violence when partners work at the same firm. We find that fewer than 2% of domestic violence incidents occur between colleagues. Nonetheless, in [Online Appendix C](#) we discuss why we include these cases in our main analysis and show that our results are robust to excluding all domestic violence cases from the sample.

The primary limitation of our measure is that we only observe incidents that are reported to the police. However, reporting is far from universal. Victimization surveys suggest that approximately 10% of physical assaults are reported to the police in Finland, with lower reporting rates for crimes considered less

serious by the victim (European Institute for Crime Prevention & Control 2009; EU Agency for Fundamental Rights 2015). This estimate is for assaults in general; there does not exist a separate estimate for between-colleague assaults. Our measure, therefore, understates the true prevalence of colleague violence and likely captures the most serious incidents. However, survey evidence suggests that reporting rates for nonpartner assault are similar for male and female victims (European Institute for Crime Prevention & Control 2009; EU Agency for Fundamental Rights 2015). This evidence suggests that the coverage of our measure might not differ significantly by gender, although we revisit this point in Section VI. If we take this 10% undercount at face value, then 3% of people in the labor force are directly involved in workplace violence as either a victim or perpetrator in Finland.⁹ As we document later, this is an underestimate of all those affected by these events, given the spillovers on the broader workforce in the firm that we uncover. However, our results on the effect of between-colleague violence on economic outcomes may not perfectly extrapolate to unreported events due to selective reporting. We discuss the interpretation of our results in light of the incomplete reporting in Section VI.

II.B. Descriptive Statistics

Table I, Panel A provides an overview of the types of crimes characterizing between-colleague violence in the police report data. In Table II, we report summary statistics for the characteristics of victims and perpetrators of between-colleague violence in the year before an incident. Table III gives the characteristics of firms in which perpetrators and victims are employed and compares their characteristics to the rest of firms in Finland.¹⁰ Between 2006 and 2019, there were more than 4,600 police reports of violence between colleagues in Finland. Table I, Panel A shows that the majority of these incidents were one of four crime types: assault (36.1%), petty assault (18.7%), menace (17.2%), and

9. According to Table II, 9,300 people are involved in between-colleague violence. If the true number of assaults is 10 times larger, this translates to 93,000 people. In 2010, 2,684,521 people were in the labor force. Assuming a majority of people are continually in the labor force over our sample period, then we find that $\frac{93,000}{2,684,521} = 3.4\%$ of those in the labor force during this time were directly involved with colleague violence.

10. Note that these tables are constructed before imposing the estimation sample restrictions in Section III.

TABLE I
CRIME TYPES

Crime types	All		Male-female		Male-male	
	Number (1)	Percent (2)	Number (3)	Percent (4)	Number (5)	Percent (6)
Panel A: Between-colleague violence						
Assault	1,678	36.1	820	41.9	625	31.5
Petty assault	868	18.7	406	20.7	275	13.8
Menace	799	17.2	338	17.3	377	19.0
Negligent bodily injury	549	11.8	93	4.7	393	19.8
Others	756	16.3	301	15.4	317	16.0
Observations	4,650		1,958		1,987	
Panel B: Noncolleague violence, male-perpetrated incidents						
Assault	233,236	54.7	84,899	48.7	148,337	58.9
Petty assault	90,275	21.2	40,208	23.1	50,067	19.9
Menace	86,165	20.2	41,690	23.9	44,475	17.7
Negligent bodily injury	16,443	3.9	7,518	4.3	8,925	3.5
Observations	426,119		174,315		251,804	

Notes. The table reports the different types of crimes committed between colleagues for the full sample, the male-female violence sample, and the male-male violence sample in Panel A, as indicated. Other contains a subset of the 57 other possible crime codes that pertain to violent crimes in Finland. See [Online Appendix A](#) for the full list of crimes, which also contains additional details on sample selection. Panel B reports the same but for noncolleague crimes, that is, crimes between two people who do not also work together. For noncolleague crimes, we restrict to the four most common types of between-colleague crime types: assault, petty assault, menace, and negligent bodily injury. In [Table II](#) we collapse the data to the individual level within a year, which is the level at which we observe individual characteristics, that is, if an individual commits multiple workplace crimes in the year he is only included once. Thus, [Table II](#) has fewer observations than this table, since here multiple crimes in the same year are all included as separate observations.

negligent bodily injury (11.8%). Panel B shows the breakdown of crime types for noncolleague violence but restricts to the four most common crime types for between-colleague violence for comparability (assault, petty assault, menace, and negligent bodily injury). The majority of crimes in these four categories are assault (54.7%), with a lower share consisting of negligent bodily injury (3.9%).

1. *Individual Characteristics.* [Table II](#), Panel A reports individual characteristics for those involved in between-colleague violence, and Panel B reports descriptive statistics for those involved in noncolleague violence but restricted to male perpetrators. Panel A shows that approximately half of the victims in between-colleague violence are women. In contrast, 84% of perpetrators are men. Panel B shows that a larger share of victims of noncolleague violence, 59%, are men. Only 4.4% of perpetrators

TABLE II
SAMPLE MEANS FOR PERPETRATORS AND VICTIMS

	All		Male-female		Male-male	
	Perpetrator (1)	Victim (2)	Perpetrator (3)	Victim (4)	Perpetrator (5)	Victim (6)
Panel A: Between-colleague violence						
Share male	0.84	0.53	1.00	0.00	1.00	1.00
Age	39.44	37.93	40.71	37.89	38.38	37.43
Share college	0.16	0.14	0.16	0.17	0.14	0.09
Share high school	0.56	0.59	0.56	0.61	0.57	0.59
Share dropouts	0.28	0.27	0.28	0.22	0.29	0.31
Employment	1.00	1.00	1.00	1.00	1.00	1.00
Earnings	38,746	32,331	38,923	27,130	40,615	36,993
Current employer tenure	5.14	4.54	5.16	4.33	5.37	4.82
Overall tenure	10.21	9.90	10.45	10.11	9.87	9.50
Share manager	0.13	0.06	0.16	0.05	0.12	0.07
Share professionals	0.21	0.18	0.23	0.21	0.20	0.14
Share clerical/service	0.17	0.22	0.17	0.28	0.11	0.12
Prior crimes	2.03	1.29	2.25	0.78	2.15	1.66
Observations	4,024	3,841	1,584	1,555	1,784	1,662
Panel B: Noncolleague violence, male-perpetrated incidents						
Share male	1.00	0.59	1.00	0.00	1.00	1.00
Age	33.03	33.95	36.30	35.92	30.72	32.50
Share college	0.04	0.07	0.05	0.10	0.03	0.05
Share high school	0.36	0.42	0.40	0.45	0.34	0.39
Share dropouts	0.60	0.51	0.55	0.45	0.63	0.56
Employment	0.35	0.40	0.37	0.43	0.33	0.39
Earnings	10,197	11,301	11,340	10,965	9,406	11,533
Overall tenure	9.22	9.22	10.04	10.20	8.65	8.54
Share manager	0.01	0.01	0.01	0.01	0.01	0.01
Share professionals	0.05	0.09	0.06	0.12	0.04	0.07
Share clerical/service	0.05	0.13	0.05	0.20	0.06	0.08
Prior crimes	9.65	3.82	9.42	2.18	9.81	4.95
Observations	426,119	426,119	174,315	174,315	251,804	251,804

Notes. Panel A reports sample means for all perpetrators and victims of colleague violence. Panel B reports the same for noncolleague violence. Data are from police reports linked to FLEED register data. In some cases there are multiple perpetrators (it is also possible but less common to have multiple victims attached to a crime code), which is why the number of perpetrators and victims are not the same. For this table, we collapse data to the individual-year level, meaning an individual who commits multiple workplace crimes in the same year only appears once, explaining the smaller number of observations compared with Table I. Income consists of all labor market earnings (the sum of wage, salary, and self-employment earnings) and is deflated to 2013 euros. Prior crimes consist of all prior police reports excluding traffic crimes and does not condition on arrests or convictions. Share college indicates those with a bachelor's degree or higher. Note that the majority of those who go on to college in Finland also earn a master's degree. High school consists of those who graduate with either a vocational-secondary or academic-secondary degree as their highest educational degree.

are repeat offenders of between-colleague violence.¹¹ Contrasting colleague with noncolleague violence in Panel B, victims and

11. This could be indicative of either a lack of repeat offenders, but could also be due to other incidents going unreported. We show that our results are robust to looking only at first offenses in Online Appendix Figure E.7.

TABLE III
ESTABLISHMENT SUMMARY STATISTICS

	Violent firms			Other firms
	All (1)	Male-female (2)	Male-male (3)	(4)
Panel A: Characteristics of the workforce				
Median number of workers	32	25	40	4
Average age of workers	39.89	39.95	39.93	41.03
Average wages	29,029.21	26,838.09	32,385.40	25,604.02
Share college	0.13	0.14	0.11	0.14
Share high school	0.57	0.57	0.58	0.56
Share dropouts	0.30	0.29	0.31	0.29
Average tenure	7.41	7.13	8.13	7.10
Turnover rate	0.23	0.25	0.23	0.31
Share of new hires	0.33	0.35	0.31	0.31
Panel B: Gender characteristics of the firm				
Share of women	0.37	0.46	0.21	0.44
Female turnover rate	0.10	0.12	0.07	0.16
Share of new hires	0.33	0.35	0.31	0.31
Share of female new hires	0.14	0.18	0.08	0.15
Average male earnings	31,779	30,257	33,905	28,758
Average female earnings	25,553	23,338	29,012	22,945
Average gender pay gap (male-female)	6,680	6,919	6,726	6,796
Median gender pay gap	5,667	5,455	6,194	4,469
Share female managers	0.22	0.26	0.14	0.30
Observations	3,154	1,520	1,357	2,632,327

Notes. The table reports sample means (unless otherwise indicated) for all establishments that experience between-colleague violence (column (1)), as well as establishments that experience male-female violence (column (2)) and male-male violence (column (3)). Sample means for all other establishments in Finland where between-colleague violence does not take place are reported in column (4). Data are from the police reports linked to FLEED administrative register data. For this table, we collapse to the yearly level, since this is the level at which we observe establishment outcomes. We also note that the smaller number of observations compared with Table II is because a single establishment can have multiple cases of workplace violence, but we only enter the establishment once per year for this table. New hires are workers who we observe in the plant in the December of that year who were not employed by the plant in December of the preceding year. Turnover rate is the share of employees leaving each year as a fraction of total employees in the plant.

perpetrators of colleague violence have significantly greater labor market attachment than those involved in noncolleague violence. Only 33%–40% of those involved in noncolleague violence are employed in the year before a crime, compared with 100% of those involved in between-colleague violence (by construction).

Table II reveals some key differences in the characteristics of between-colleague male-female and male-male violence that go beyond simply victim gender. On average, victims and perpetrators of male-male workplace violence are the same age and have similar incomes. Male-male crimes are slightly more likely to have multiple perpetrators compared with male-female crimes, where almost all crimes consist of a single perpetrator. In

contrast, victims of male-female workplace violence are younger and earn €11,793 less per year than their male perpetrators, equivalent to 30% lower incomes. On average, the income gap between female victims and male perpetrators is almost twice as large as the average gender pay gap within the firm (see [Table III](#)). In addition, male perpetrators of male-female workplace violence are approximately three times as likely to be in management, using the (coarse) occupational codes, compared with their female victims. Thus, victims of male-female violence are relatively low-earning women in the firm compared to their perpetrators, who are relatively high-earning men. This is not the case for male-male workplace crimes, which are more likely to occur between relative equals in the firm. We do not see the same large difference in average incomes for male perpetrators and female victims in the case of noncolleague violence. [Table II](#), Panel B highlights that the perpetrator and victim have almost identical earnings for both male-male and male-female noncolleague crimes. Thus, it seems that male-female between-colleague violence is uniquely characterized by the fact that women are attacked by an individual in a position of economic power.

The types of crimes that characterize male-female between-colleague violence are also more serious than those for male-male violence. This could be indicative either of differences in underlying severity by victim gender or a higher threshold for reporting in the case of male-female crimes. [Table I](#) shows that 63% of male-female incidents are assaults, compared with 45% of male-male incidents. Beyond assault, the next most prevalent crime for male-male workplace violence is negligent bodily injury, which is characterized by a lack of care, while for male-female violence it is menace, which requires a perpetrator intentionally causing fear of serious injury or death.¹² Note that [Table I](#) reports statistics at the case level, and thus has more observations than [Table II](#), which collapses the data to the individual level in a year, that is, if a person commits multiple workplace crimes in the year, they are only included once in [Table II](#).

12. The greater severity of male-female crimes is also apparent if we look at the rate at which these types of crimes are incarcerated. A much higher share of menace and assault cases in court result in prison sentences (9.5% and 7.3%, respectively), compared to negligent bodily injury (0.6%). These incarceration rates are for all cases with the relevant crime code and do not restrict to cases of between-colleague violence.

Given these differences, we largely analyze male-female violence and male-male violence separately in the rest of the article. The economic relationship between victim and perpetrator, and the severity of the crime reported, indicate that these are likely to have very different effects on victims and the wider firm. We also note that for the main analysis, we do not single out cases with a female perpetrator for two reasons. First, women are rarely perpetrators. In only 16% of cases do we observe a female perpetrator of workplace violence. Furthermore, in many cases involving a female perpetrator, she is also recorded as a victim, meaning these are often not clear-cut cases of female-perpetrated violence (see [Online Appendix Figure E.1](#)). There are fewer than 250 clear-cut cases of female-male violence and female-female violence. In contrast, the vast majority of male-female and male-male cases have a clear-cut perpetrator who is not also classified as a victim in the same incident. Thus, small sample sizes make it difficult to say anything conclusive about female-perpetrated crimes. We report estimates for these crimes in [Online Appendix B](#).

2. *Victim and Perpetrator Earnings Growth Prior to Violence.*

A reasonable hypothesis before examining the data is that individuals who are attacked by a colleague may have been victimized for many years leading up to the event that they report to the police, resulting in a deterioration in earnings in the years preceding the event. Similarly, years of harassing a colleague might lead to a deterioration in earnings for perpetrators prior to the reported event. This is not what we find. In [Figure I](#), we graph earnings of victims and perpetrators compared to all workers in Finland of the same gender and of a similar age and education level.¹³ We see that the pretrends are identical, with no evidence of differences in earnings trends prior to the violent incident. This growth in earnings leading up to the event provides a stark

13. [Online Appendix Figure E.4](#) gives employment growth in the years before the event (i.e., for years -5 to -1). There is robust employment growth in the years before a violent incident for both victims and perpetrators. Although this employment growth is somewhat mechanical (the victim and perpetrator must work together the year before the event for it to be violence between colleagues, so even if there is unemployment in prior years, by year -1 they must be employed), the earnings growth is less mechanical. While moving from unemployment to employment will mechanically increase earnings, those who are continually employed could have experienced decreases in their earnings prior to the event.

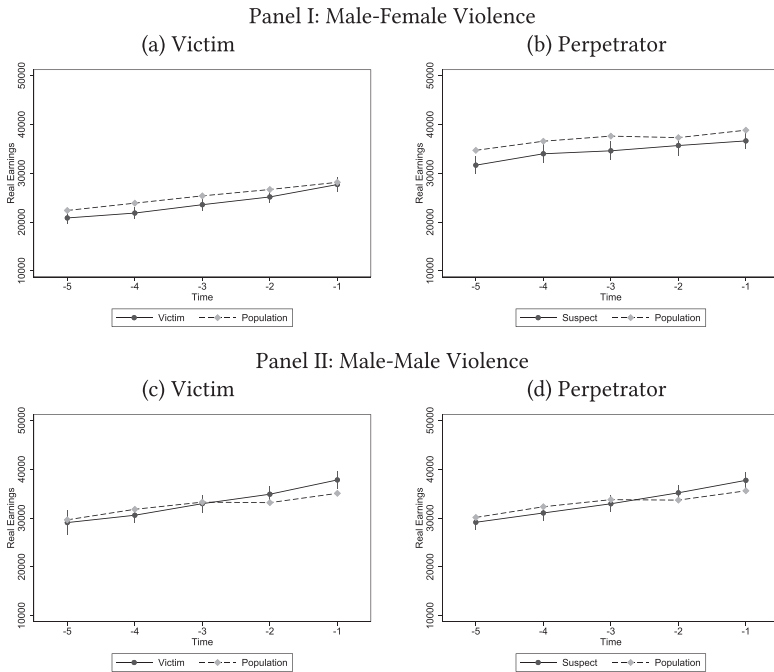


FIGURE I

Earnings Growth of Victims and Perpetrators over the Five Years Prior to Violence Compared with Earnings Growth of All Workers in Finland

Each figure reports the earnings in the five years prior to the violent incident for the victim (left side) or perpetrator (right side), relative to all workers in Finland of the same gender, and of a similar age and education level. To calculate all Finnish workers' earnings' pretrend, we assign 2010 as the pseudo-year of the event for all Finnish workers and calculate their previous five years of earnings. The first row reports trends for male-female violence. The second row reports trends for male-male violence. Earnings consist of all labor market earnings (the sum of wage, salary, and self-employment earnings) and is deflated to 2013 euros. The *x*-axis measures years relative to the year in which violence takes place. We reweight the outcomes of all workers in Finland to match the age and education distribution of victims and perpetrators.

contrast to the abrupt deterioration in labor market outcomes that we document after the violent incident.

Figure I also makes the difference in the economic relationship between victims and perpetrators of male-male versus male-female violence even clearer. For male-male violence, earnings levels and growth before the incident are indistinguishable between perpetrators and victims. For male-female violence, the

women make significantly less on average than their male perpetrators. In [Online Appendix Figure E.3](#), we report similar results to [Figure I](#) but looking only at other workers in the firm. The dotted line again shows the earnings growth of victims (left panel) and perpetrators (right panel), but now the solid line shows the earnings growth of other workers in the same establishment of the same gender, and of a similar age and education level. We see that women who are attacked are relatively low-earning women in the establishment, even relative to other women of the same age and education. However, we still observe parallel trends in earnings growth before the event.

3. Our Data and Sexual Harassment. Our data capture all cases of violence between two colleagues reported to the police. While this article is concerned with between-colleague violence in general, we are also interested in the extent to which our data captures sexual harassment and #MeToo type cases. Although our data are unlikely to include inappropriate sexually motivated comments, it will include groping and stalking incidents, which are commonly cited forms of sexual harassment, as well as rape and sexual violence.¹⁴

Groping incidents were categorized as assaults or petty assaults in the Finnish crime code prior to 2015, and our main analysis includes only events up to 2014 so that we can observe outcomes five years postevent. Assault/petty assaults make up the vast majority of our male-female cases, as shown in [Table I](#). [Table I](#) also shows that while the first two most common between-colleague crimes for male-female violence are assault and petty assault, the third most common category is menace. This consists of “a person who points a weapon at another or otherwise threatens another with an offense under such circumstances that the person so threatened has reason to believe that the personal safety or property of himself or someone else is in serious danger.” (translated from Section 25 of the Finnish Criminal Code) Stalking was added to the Finnish criminal code in January 2014. Prior to 2014, stalking involving threatening behavior would have fallen under the menace crime code ([Hossa 2012](#)).¹⁵

14. Groping was a particularly common type of #MeToo case. For example, Andy Dick, Paul Rosenthal, Al Franken, Raul Bocanegra, Dean Westlake, and many more high-profile men were accused of groping during #MeToo.

15. If stalking behavior is threatening, it would be classed as a menace before and after 2014. See <https://poliisi.fi/en/stalking>. In some cases, stalking behavior could have also fallen under the coercion code before 2014.

Our data also include incidents categorized as rape, aggravated rape, and sexual abuse (see [Online Appendix A.4](#)). However, the Finnish crime code had a very strict definition for rape and sexual violence over the period covered by our data. It did not, for example, include a consideration of consent and was rather based on the degree of force and physical violence perpetrated. This approach has been internationally criticized, and experts have argued that it caused many acts of sexual violence not to be labeled as such, leading to perverse outcomes at other points of the criminal justice system.¹⁶ For example, the official baseline report by the Council of Europe on the consistency of Finnish policy with its *Convention on Preventing and Combating Violence against Women and Domestic Violence* noted that the Finnish criminal definition of rape created “higher thresholds of evidentiary standards of physical resistance” and “that not all forms of sexual violence are criminalised [as such] in Finland” ([Council of Europe 2019](#)). For this reason, and because our interest is in violence between colleagues more broadly, we do not restrict our analysis only to cases categorized by police as sexual violence and rape.

We also note that many anecdotal accounts of sexual harassment at work are characterized by men in positions of power taking advantage of their (relative) power to attack lower-ranked women with few repercussions. These power differences are already apparent in the descriptive results for male-female violence in [Figure I](#), and we investigate the important role of power dynamics in detail in [Section III.C](#).

4. *Establishment Characteristics.* Turning to establishment characteristics, [Table III](#) shows that establishments in which colleague violence is reported do not appear to be negatively selected relative to all other establishments in Finland. They have higher wages and longer tenure on average. Establishments where male-female crimes take place also do not appear to be bad places to work for women based on observables. They have a higher female share overall, a higher share of female new hires, similar gender pay gaps compared to male-male violent establishments, and

16. For example, Finland’s supreme court ruled in 2018 that a man who sexually abused a 10-year-old girl could not be charged with rape as there was no evidence the crime had involved violence or that the child was overcome by fear ([Boffey 2019](#)).

approximately the same female share and gender pay gaps as all other establishments in Finland. The dimension along which violent and nonviolent establishments differ the most is firm size, as violent establishments are significantly larger. This may be partly mechanical: in larger establishments, there are more potential combinations of colleagues between whom violence can occur. This means that even if the assault risk per worker is still the same, the probability that an event occurs in a larger establishment is higher.

In [Online Appendix](#) Table D.1 we report estimates from a simple linear probability model with a dummy for workplace violence as the outcome in column (1) and the occurrence of male-female and male-male violence as separate outcomes in columns (2) and (3). Establishments in the public sector, administration, and manufacturing are all significantly more likely to experience both male-male and male-female workplace violence relative to other industries ([Online Appendix](#) Figure E.2). Mining and quarry industries are more likely to experience male-male violence, but not more likely to experience male-female violence. However, the apparently higher rate of workplace violence in public sector firms may be an artifact of better reporting or support structures in those firms. Although analysis of what causes reporting is beyond the scope of this article, it is an important area for future research. Controlling for industry fixed effects, we see that establishments in which colleague violence occurs are slightly younger, less educated, and lower paying, although the magnitudes of these coefficients are all small. We also find that establishments in which colleague violence occurs are less likely to be female-managed, especially in cases where male-female violence takes place. There is no statistically significant difference in turnover rates nor in gender pay gaps between violent and nonviolent establishments.

III. EFFECTS ON VICTIMS AND PERPETRATORS

III.A. Descriptive Labor Market Outcomes

[Figure II](#) gives the raw change in employment rates of perpetrators and victims the year after an incident relative to the year before. There is a large and immediate drop in employment at the exact point of the incident for perpetrators and victims of both male-female and male-male violence. The magnitude of

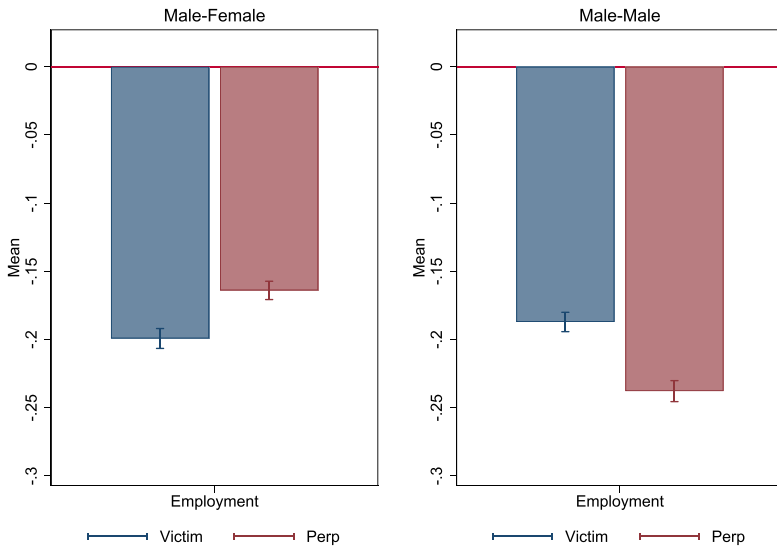


FIGURE II

Asymmetry in the Raw Effects of Between-Colleague Violence on Employment and Earnings

The figure reports the change in employment one year postviolence compared to the year before violence. The left graph reports this raw employment change for male-female violence for victims (in the blue bar on the left; color version available online) and perpetrators (in the red bar on the right). Similarly for male-male violence in the right graph. Ninety-five percent confidence intervals for the mean change in employment are depicted in whiskers. The employment measure we use is an indicator for whether an individual was employed in the last week of the year (the reference week).

this decline varies according to the gender of the victim. Female victims experience a 20 percentage point decline in their employment, while their male perpetrators experience a smaller decline in employment of just over 15 percentage points the year after the event compared to the year before. Male perpetrators of male-male violence experience a 24 percentage point decline in employment compared with an 18 percentage point decline for their victims. Note that these sudden declines follow consistent and robust increases in employment rates and earnings in the years leading up to the violent incident for both perpetrators and victims (see [Figure I](#), and [Online Appendix Figures E.4 and E.5](#)).

III.B. *Estimated Labor Market Outcomes*

The raw means depicted in [Figure II](#) suggest potentially large effects of between-colleague violence and an important asymmetry in those impacts for male-female versus male-male events. The main concern with these descriptive results is that there may be some other factor that explains the immediate drops in employment following the violence. For example, many workers naturally separate from employment each year. Moreover, perpetrators might target economically vulnerable colleagues, or might themselves be struggling in the firm and lashing out as a result. The descriptive results above would fail to recover the true effects of violence if declines in a victim's (or perpetrator's) employment directly after the event occur not because of the violence but because the victim (or perpetrator) was going to be fired or quit that year in any case.

To mitigate this type of concern, we use a matched difference-in-differences design, comparing victims and perpetrators of workplace violence relative to a matched control observation ([Goldschmidt and Schmieder 2017](#); [Aneja and Xu 2022](#)). This approach allows us to carefully compare the evolution of outcomes before and after the incident for treatment and control observations with similar characteristics.¹⁷ Likewise, we estimate the effect of workplace violence on firm outcomes relative to a matched control firm later.

Formally, we find a victim's (and perpetrator's) nearest-neighbor match on the basis of their age, education level, gender, managerial status, and employment and income history in the five years before the incident.¹⁸ We restrict to victims and perpetrators that we can follow five years before until five years after the event. For firm outcomes, we find the nearest-neighbor match on firm characteristics: firm size, turnover rate, industry, the average age of workers, average education of workers, share of new

17. We employ a stacked difference-in-differences design comparing never treated to treated individuals which addresses recent concerns in this literature. See [Cengiz et al. \(2019\)](#) for more details.

18. In the matching, we control for age and log income linearly and control for education level, gender, managerial status, and employment by fixed effects. When there are cases of zero income, we use $\log(\text{income}+1)$. Note that we only make this adjustment in the matching. When reporting impacts on income, we report effects on future labor market income as a share of average income in the five years prior to the incident. For perpetrators, we also control for the number of cumulative crime records linearly.

hires, and gender composition.¹⁹ These exercises leave us with a match for the victim, a match for the perpetrator, and a match for the firm that all appear identical on observables in the five years prior to the year in which violence occurs but who do not experience workplace violence.²⁰

With matched control and treatment observations in hand, we estimate the following regression model:

$$(1) \quad Y_{ibt} = \sum_{j=-5, j \neq -1}^5 \delta_j D_{ib,t-j} + \alpha_{ib} + \gamma_t + \omega_j + Age_{ib} \times \beta_j + \epsilon_{ibt},$$

where Y_{ibt} represents the outcome of interest for victim (perpetrator) i in base-year sample b at time t .²¹ b is the year in which the violent incident occurs. $D_{ib,t-j}$ is an indicator variable for the treatment (workplace violence) separately for each year j since the event. δ_j are the coefficients of interest, identifying the effects of the violent incident on victim, perpetrator, or firm outcomes relative to the matched counterfactual. We omit the year prior to the event ($j = -1$), which means that all estimates of δ_j are relative to the year before the incident. In addition, we include individual-incident-year fixed effects (or firm-incident-year fixed effects when examining firm outcomes), α_{ib} , year fixed effects, γ_t , time since crime fixed effects, ω_j , and age at the time of the incident by time since event interactions, $Age_{ib} \times \beta_j$.²² Standard errors are clustered at the individual level (or firm level when examining firm outcomes).

We report the separate yearly effects δ_j for the five years after the incident or report overall difference-in-differences (DiD) estimates for our outcomes of interest. DiD estimates provide the differences in outcomes five years after versus five years before for victim, perpetrator, or firm outcomes relative to their matched counterfactual. Comparisons always occur between treated and

19. We find similar results for firm outcomes whether we use the individual match or the firm match.

20. We do not restrict to same-firm matches as they are also treated given the colleague spillovers we document.

21. When we examine firms, Y_{ibt} represents a variety of firm-level outcomes, such as headcount, exit, and workforce composition, for firm i in base-year sample b at time t .

22. People are different ages at base year, so this is not collinear with individual and time-since-event fixed effects. Our results are robust to adding incidence-year-specific time-since-event fixed effects.

never-treated individuals to address concerns of bias in event study estimates (Goodman-Bacon 2018; Sun and Abraham 2021), that is, this is a stacked DiD exercise as in Cengiz et al. (2019).

A key assumption for a causal interpretation of these results is that the employment of the victim (perpetrator) would have evolved along common trends as their matched control in the absence of one colleague attacking another. A similar assumption must hold for firms and their matched controls. Online Appendix Figures E.4 and E.5 show raw employment and income before and after the violent incident for victims and perpetrators, along with their matched controls. Prior to the incident, the raw income and employment for both the victims and their matches, as well as the perpetrators and their matches, are identical. There is a sharp discontinuity in the labor market outcomes of the perpetrators and the victims after a violent incident, with employment and income dropping substantially after workplace violence. We do not see these effects for the control observations, aside from a small mechanical drop in employment due to natural separation (we have required employment in a firm in year -1). In addition, in Section III.D we present a placebo check showing that matched controls and treated individuals' employment do evolve similarly in the post period when we instead examine a period where no violence takes place, providing additional reassurance for our approach. We discuss remaining concerns for the validity of our estimates and how we address them in Section III.D, after presenting the main estimates.

1. Results: Full Effects of Between-Colleague Violence.

Figure III gives the event study coefficients of interest from equation (1) for victims and perpetrators with employment as the outcome of interest in the first two rows, and income in the last two rows. Figure IV reports the aggregated difference-in-differences results. In Figure III, Panels I and II we see that the employment rate of victims of male-female violence falls by approximately 8.2 percentage points in the first year after the incident relative to their matched controls, which grows to a 12.1 percentage point fall in employment by five years later. There are similarly large effects on labor market incomes, with victims losing over 20% of their incomes relative to their average incomes before the incident (Figure III, Panels III and IV). Perpetrators of male-female workplace violence also experience negative effects

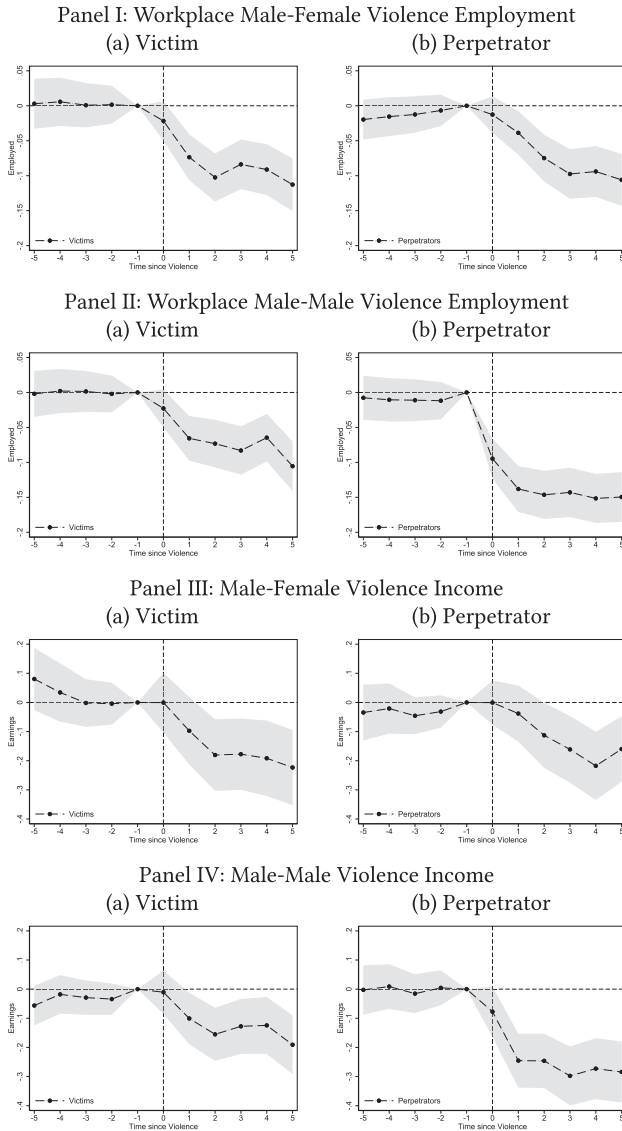


FIGURE III

Effect of Workplace Violence on Employment and Incomes of Victims and Perpetrators

FIGURE III

(Continued) The figure reports effects of between-colleague violence on the victim (left side) or perpetrator (right side). The first (third) row reports effects of male-female violence on employment (income), and the second (fourth) row reports similarly for male-male violence. The estimates use the matched control to identify effects five years before and five years after a violent incident against a colleague (see [equation \(1\)](#)). Employment indicates whether the individual was employed in the last week of the year. Income is total taxable labor market income (the sum of salary, wages, and self-employment earnings), and we estimate effects measured as all taxable income at the end of the year as a fraction of the total income in the year before the incident. Standard errors are clustered at the individual level.

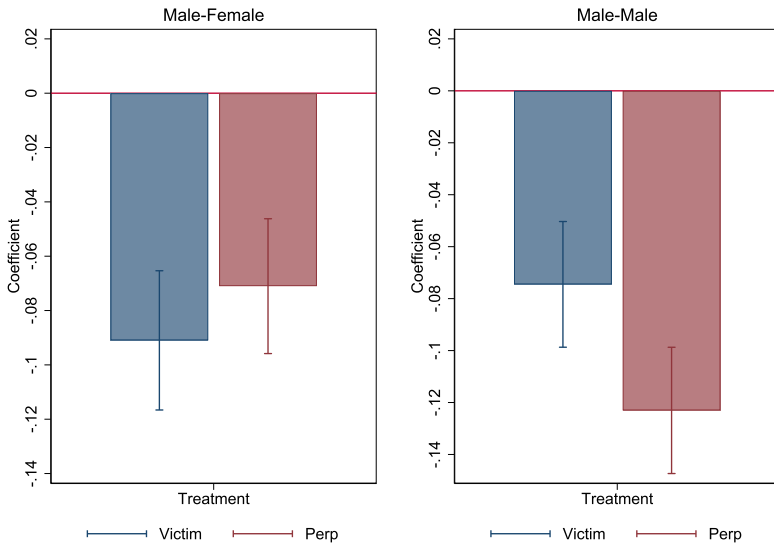
on their employment and incomes.²³ The second and fourth rows of [Figure III](#) report event study coefficients for victims and perpetrators of male-male violence. Victims see an immediate 7.1 percentage point drop in employment that grows to 11.7 percentage points five years later, and their incomes decrease by just under 20%. Perpetrators see an immediate employment decline of 14.3 percentage points that grows to 15.2 percentage points five years later.

The event studies show that workplace violence has large economic consequences for victims and perpetrators. They also demonstrate a striking asymmetry in the labor market effects of between-colleague violence for victims and perpetrators across male-female and male-male incidents, consistent with the raw descriptive results. [Figure IV](#) succinctly summarizes this, showing the overall DiD effects on the perpetrators and victims of male-female versus male-male crimes.²⁴ For male-male violence, perpetrators suffer a significantly greater labor market cost than their victims: their employment rates fall by over 12 percentage points in the five years after an incident, and this effect is significantly different than the 7.5 percentage point decline in employment for victims. This is the result one might expect. It seems reasonable that perpetrators of workplace violence experience more negative outcomes after attacking a colleague than their victims do. In particular, we might expect the firm to punish perpetrators after the incident, including possibly firing the perpetrator.

23. In the five years after the incident, there is a 7.1 percentage point drop in perpetrator's employment and an 11.6% fall in income relative to the prevalence baseline (see [Online Appendix Table D.3](#)).

24. Results are similar if we only estimate results one year or three years post, see [Online Appendix Figure E.6](#).

(A) Employment



(B) Labor Market Income

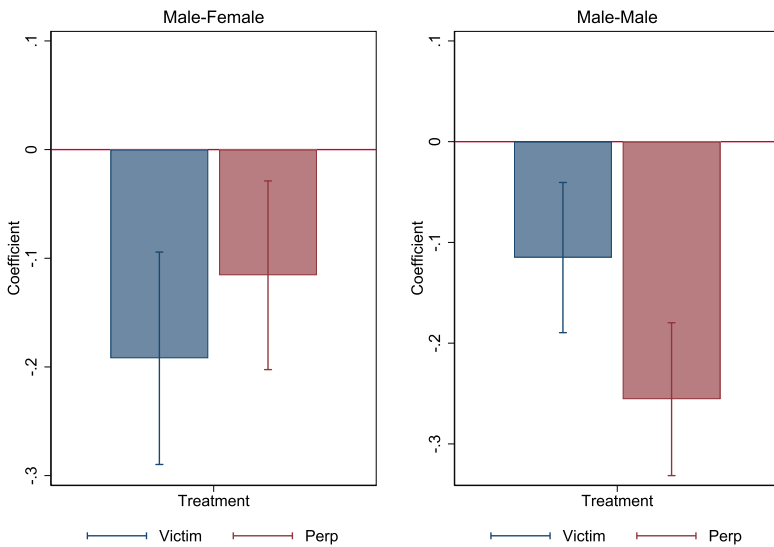


FIGURE IV

Asymmetry in Employment and Earnings Impacts of Workplace Violence

FIGURE IV

(Continued) The figure reports estimates of δ_t obtained using equation (1), where we collapse into a pre- and postperiod to recover difference-in-differences estimates. The left figure reports DiD estimates for male-female violence for victims (in the blue bar on the left) and perpetrators (in the red bar on the right). Similarly for male-male violence in the right figure. Panel A reports effects on employment, while Panel B reports effects on labor market income. Ninety-five percent confidence intervals are depicted in whiskers around the estimates. Employment indicates whether the individual was employed in the last week of the year. Income is total taxable labor market income (the sum of salary, wages, and self-employment earnings) after the event as a fraction of the average total income in the five years prior. Standard errors are clustered at the individual level.

In contrast, there is no statistically significant difference in the employment effects of male-female violence between perpetrators and victims. Indeed, if anything, perpetrators experience smaller labor market consequences compared with their victims. Overall, perpetrators' employment rates fall by 7.1 percentage points. This is smaller than, but not statistically distinguishable from, their female victims, whose employment falls 9.1 percentage points. Panel B shows a similar asymmetry in the effect of workplace violence on the percent earnings losses of victims versus perpetrators for male-female violence versus male-male crimes, (see also [Online Appendix Table D.4](#)).²⁵

In [Online Appendix Figure E.8](#) we examine the effect of a violent incident on leaving the firm rather than employment. Leaving the firm occurs if the victim/perpetrator moves to unemployment, but also if they leave for a different firm but remain in employment. We find a similar asymmetry in the perpetrator effects of between-colleague violence. Male-male perpetrators are just over 10 percentage points less likely to be employed in the same firm in the five years following a violent incident. In comparison, perpetrators of male-female violence are just over 5 percentage points less likely to remain in the same firm relative to their matched control, almost half the size of male-male perpetrators. Victims of male-female violence are, consistent with our employment effects, less likely to remain in the same firm, although the difference with male-male victims is not statistically significant.

25. Victims of male-female crimes experience a loss in income of €2,720 on average (see [Online Appendix Table D.4](#)). This is a sizable magnitude: [Online Appendix Table D.4](#), column (3) shows that female victims' incomes fall 19% on average compared to the previolence baseline.

2. *Results: Comparison to Noncolleague Violence.* The foregoing estimates focus on the effect of being attacked by a colleague relative to a counterfactual in which no violent event occurs. However, an equally interesting question is how the effect of being attacked by a colleague compares to the effect of being a victim of a similar crime, but with a perpetrator who is not a colleague.²⁶ Similarly for perpetrators. For firms, we might be interested in whether there is a difference in outcomes when an employed perpetrator attacks a colleague versus if they commit a crime against a noncolleague.

As shown in Table II, individuals involved in crimes with noncolleagues have very different characteristics than those involved in between-colleague crimes. Thus, to better isolate the effect of crime between colleagues versus crimes between noncolleagues, we carefully compare similar individuals across the two groups. Specifically, we match exactly on employment status in the year preceding the crime, crime type, and victim and perpetrator gender. We identify an individual's nearest neighbor on the basis of their age, education, and employment/income outcomes in the five years before an incident.²⁷ In other words, we carefully compare outcomes of victims and perpetrators before versus after violence, comparing observationally identical people where the only difference is that the perpetrator (or victim) attacked (or was attacked) by someone within versus outside of their workplace.

The results of this exercise are presented in Figure V, which gives the matched DiD estimates for victims and perpetrators of male-female and male-male violence with the noncolleague violence counterfactual. The limited employment losses for perpetrators of male-female workplace violence are even more pronounced when one considers the counterfactual of violence between noncolleagues. For male-male violence, we see no significant differences between the employment losses for men who attack colleagues compared with observationally equivalent individuals who commit the same category of crime with a non-colleague, and the same is true for male victims. These results

26. Related is a small literature showing negative effects of crime in general on victims in other contexts (Johnston, Shields, and Suziedelyte 2018; Koppensteiner and Menezes 2021; Bindler and Ketel 2022; Currie, Mueller-Smith, and Rossin-Slater 2022).

27. For this exercise, we restrict our analysis to the four most frequent crime codes in our data (Table I) to allow an exact match on crime type.

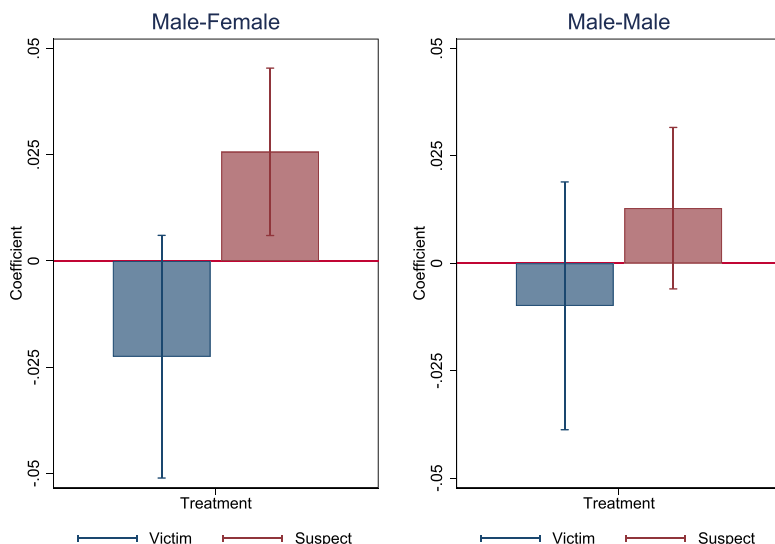


FIGURE V

Comparing Workplace and Nonworkplace Violence Effects on Employment

The figure reports estimates of δ_i obtained using [equation \(1\)](#), where we collapse into a pre- and postperiod to recover difference-in-differences estimates. However, unlike the main estimates, which use a nearest-neighbor match who is not also a victim (or perpetrator when estimating perpetrator impacts) of a crime, here we compare outcomes to a nearest-neighbor match who was also a victim (or perpetrator) of one of the same main types of crimes in [Table I](#), but where the two did not work in the same plant and so were not colleagues. The left figure reports DiD estimates for male-female crimes for victims (in the blue bar on the left) and perpetrators (in the red bar on the right) compared with the effects for nonworkplace victims and perpetrators. Similarly for male-male crimes in the right figure. Ninety-five percent confidence intervals are depicted in whiskers around the estimates. Employment is measured at the end of the year. Standard errors are clustered at the individual level.

suggest that male-male violence has similar effects whether it occurs within or outside of the firm.

In contrast, for male-female violence, victims of between-colleague violence suffer 2.3 percentage points greater employment losses compared with women who are observationally equivalent and victims of the same crime but where the perpetrator was not a colleague (p -value = .122). In contrast, perpetrators of colleague violence who attack women suffer significantly smaller employment losses. Perpetrators are 2.5 percentage points more likely to keep their jobs over the five years following the incident

when they attack a workplace colleague compared to an observationally equivalent individual who commits a crime against a noncolleague (p -value = .011). This is a striking finding given that it is plausible that violence against a colleague is more likely to be public to decision makers in the firm.

III.C. The Role of Power Discrepancies between Victim and Perpetrator

A key takeaway from the descriptive results and both counterfactuals is that men who attack female colleagues are comparatively insulated from employment costs compared to their victims, men who attack male colleagues, and men who attack women outside the firm. In this section, we explore a possible explanation for this pattern: the economic standing of perpetrators relative to victims in the firm. Anecdotal evidence from high-profile media accounts of male-female violence suggests that inequality between victims and perpetrators, and being attacked by an individual in a position of power, is especially problematic for victims. When describing the aftermath of her assault at the hands of Harvey Weinstein to her *New York Times* editorial, Rowena Chiu wrote “Harvey was a power player, and I was the lowest person on the totem pole. Assistants are the unseen work force that props Hollywood up, and yet we have zero leverage. I was invisible and inconsequential” (Chiu 2019).²⁸ This was a common theme for female victims attacked by male colleagues, even when the perpetrators were not as famous and powerful as Weinstein.²⁹ Thus, it seems that having perpetrators in a position of power might play an important role in how these events are adjudicated and the eventual effects on victim and perpetrator outcomes.

28. Widespread media coverage of these events did eventually lead to repercussions in some cases. Most notably, Weinstein was convicted and sentenced to prison in 2020 after a five-day deliberation. The first police report against Weinstein occurred in 2015 and was quickly dismissed after a two-week investigation.

29. For example, a Starbucks worker commented to the Huffington Post, “I worked at Starbucks for three years during college. When one of my coworkers and I reported our shift manager for sexual harassment we were told we’d be laid off the following week” (Agrawal 2015).

TABLE IV
THE ROLE OF POWER DISCREPANCIES BETWEEN VICTIM AND PERPETRATOR

Dependent variable:	Victim employment		Perpetrator employment	
	(1)	(2)	(3)	(4)
Panel A: Male-female				
Treatment × Perp is manager	−0.070 (0.034)		0.060 (0.022)	
Treatment × Income gap		−0.037 (0.019)		0.086 (0.019)
Treatment	−0.085 (0.013)	−0.073 (0.015)	−0.077 (0.013)	−0.114 (0.017)
Observations	22,861	22,861	23,253	23,253
Dependent variable mean	0.854	0.854	0.867	0.867
Panel B: Male-male				
Treatment × Perp is manager	−0.063 (0.043)		0.137 (0.023)	
Treatment × Income gap		−0.066 (0.021)		0.125 (0.019)
Treatment	−0.070 (0.012)	−0.049 (0.014)	−0.134 (0.013)	−0.172 (0.016)
Observations	23,076	23,076	23,374	23,374
Dependent variable mean	0.857	0.857	0.861	0.861
Year fixed effects	✓	✓	✓	✓
Time since crime fixed effects	✓	✓	✓	✓
Individual-incident-year fixed effects	✓	✓	✓	✓
Age × Time since crime	✓	✓	✓	✓

Notes. The table reports difference-in-differences estimates from equation (1) collapsed into a pre- and post-period. In all cases, the dependent variable is employment measured at the end of the year. Panel A reports estimates for only male-female workplace crimes, and Panel B report estimates for male-male workplace crimes. Data are from police reports linked to FLEED register data. Manager is defined from the occupation variable in the data. Income gap is the difference in the income rank in the firm of the perpetrator and their victim. Standard errors are clustered at the individual level.

1. *Power Differentials: Main Results.* In Table IV we consider heterogeneity in the effects of workplace violence by the economic standing of victims relative to perpetrators in the firm.³⁰ First, we interact the “treatment,” that is an attack by a colleague, with whether the perpetrator is a manager for male-female crimes in Table IV, column (1) and for male-male crimes in column (3).³¹ Second, we interact the treatment of workplace violence with the difference in the income rank in the firm of the perpetrator and their victim, a broader measure of economic power

30. In Online Appendix Table D.5, we consider heterogeneity just by individual characteristics rather than differences across victims and perpetrators. Few significant patterns emerge.

31. This exercise uses the coarse occupation variable in the data and whether this variable indicates a managerial role for the perpetrator.

differences between victim and perpetrator. This approach is possible because we observe the universe of workers in the firm and their labor market earnings, in addition to the perpetrator and his victim. For example, the income rank gap between a perpetrator at the 75th income percentile in the firm and a victim at the 25th income percentile would be 0.5. We report the results of this second exercise in [Table IV](#), columns (2) and (4).

[Table IV](#) shows that the economic power of perpetrators matters for the consequences of male-female and male-male violence. In both cases, victims face greater employment losses when perpetrators are managers and there is a greater income gap between the victim and perpetrator. For perpetrators, the effect is the opposite: their employment rates are less severely affected when they occupy relative positions of power in the firm. For male-female (male-male) crime, victims' employment rates fall by 7 (6.3) percentage points more when their perpetrator is a manager. However, perpetrators who are managers are 6 (13.7) percentage points less likely to be unemployed in the five years following an incident. Thus, power in the firm allows perpetrators to avoid more severe consequences, while their victims experience worse outcomes.

These results do not admit an interpretation that female victims are simply more harmed by and/or less effective at holding powerful perpetrators to account than male victims are: there are insignificant differences in the treatment effect of workplace violence between male and female victims. However, these results could partially account for the limited employment effects on male perpetrators of male-female crimes where such economic inequality between the victim and perpetrator is more common (see [Section II.B](#)).

2. Power Differentials When Using the Nonworkplace Violence Counterfactual. Next we examine if the power differential explanation also holds true when we use the counterfactual of nonworkplace violence. We report results in [Online Appendix Table D.7](#) that replicate [Table IV](#) but replace the counterfactual of no violence with the counterfactual of nonworkplace violence. We find that it is not simply being higher status in society that matters, but rather being high status when the colleague is also in the firm that matters for how these events affect employment.

Specifically, for male-female (male-male) crime, victims' employment rates fall by 4 (3.3) percentage points *more* when their

perpetrator is a manager compared with nonworkplace victims whose perpetrators are also managers, but not in the same firm. However, perpetrators who are managers are 4.1 (7.2) percentage points *less* likely to be unemployed in the five years following an incident compared with nonworkplace perpetrators.

3. *Gender, Power Differentials, or Both?* The previous sections showed that for both male and female victims, having a manager attack them leads to much worse outcomes, and in both male-female and male-male violence, perpetrators who are managers or make significantly more than their victims are relatively insulated from unemployment. In this section, we explore what this means for our interpretation of the asymmetry in effects across male-female and male-male crimes documented in Figure IV.

The descriptive statistics in Section II.B showed that female victims of between-colleague violence are lower status in the firm and are more likely to be victims of more economically powerful men. This suggests an important composition difference between male-female and male-male between-colleague violence. Figure VI formally captures this point. It reports estimates from three linear probability models where the dependent variable is: (i) the managerial status of the perpetrator; (ii) whether the perpetrator is in the top 20% of earners in the firm; (iii) whether the perpetrator is in the bottom 20% of earners in the firm. The independent variable is the gender of the victim, with controls for the crime type. The red bar (color version available online) indicates that female victims are significantly more likely to have a perpetrator who is a manager, and the blue middle bar indicates she is significantly more likely to be attacked by a manager in the top 20%. The green bar shows that female victims are significantly less likely to be attacked by a perpetrator in the bottom 20%.

In Online Appendix Figure E.9 we show that female victims are more likely to be in the bottom 20% in terms of income in the workplace compared with male victims. This could be due to women being of lower status on average in the firm. To explore this possibility, we also estimate the effect of a victim being female on the within-firm within-gender income rank of the victim, constructing an indicator for whether the woman is in the bottom 20% of female earners in the firm. We still find a significant negative relationship between the within-gender income rank of victims and whether the victim is a woman. Thus, compared with

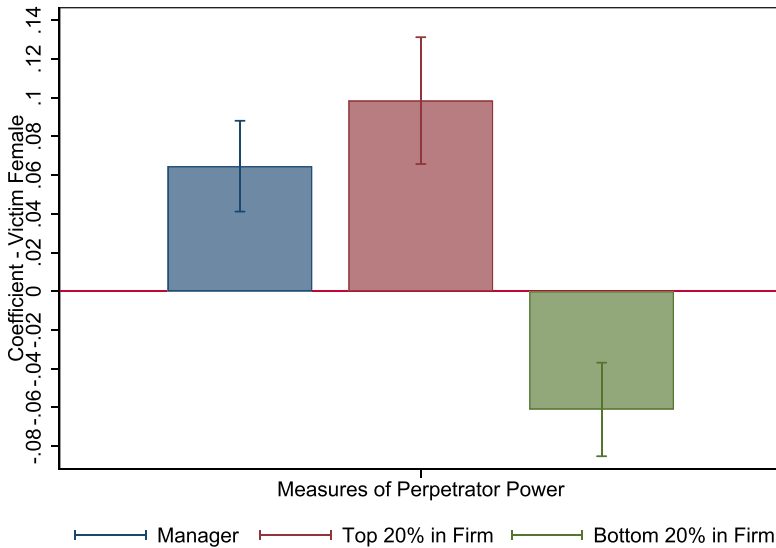


FIGURE VI

Perpetrators Who Attack Women Are More Powerful in the Firm

The figure reports descriptive estimates of the correlation between victim gender and the power of the perpetrator, focusing on whether the perpetrator is a manager (in the blue, leftmost bar), whether the perpetrator is in the top 20% of earners in the firm (in the red, middle bar), and whether the perpetrator is in the bottom 20% of earners in the firm (in the green, rightmost bar). Ninety-five percent confidence intervals are depicted in whiskers.

male victims, female victims are relatively low earners in the firm compared to all workers and their gender peers.³² Thus, while power differentials matter for male and female victims (and also for perpetrators who attack men or women), women are more likely to be attacked by someone in power and be lower earning in the workplace.

We analyze the degree to which these economic power imbalances can “explain” the gaps in perpetrator outcomes for male-female versus male-male violence in Table V. We pool observations from male-female and male-male violence and interact the treatment of between-colleague violence with the gender of the

32. This is also consistent with the fact that female victims on average have lower tenure in the firm than male victims.

TABLE V
POOLED EMPLOYMENT EFFECTS AND POWER DISCREPANCIES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Dependent variable: victim employment							
Treatment	-0.0746 (0.0123)	-0.0679 (0.0135)	-0.0653 (0.0135)	-0.0501 (0.0139)	-0.0736 (0.0104)	-0.0560 (0.0111)	-0.0500 (0.0139)
Treatment × Female victim	-0.0157 (0.0180)	-0.0176 (0.0180)	-0.0161 (0.0180)	-0.0119 (0.0181)			-0.0114 (0.0181)
Treatment × Perp is manager			-0.0640 (0.0272)		-0.0646 (0.0272)		-0.0478 (0.0285)
Treatment × Income gap				-0.0499 (0.0140)		-0.0506 (0.0139)	-0.0445 (0.0146)
Treatment × Assault		-0.0153 (0.0144)	-0.0101 (0.0143)	-0.0126 (0.0144)	-0.00904 (0.0143)	-0.0117 (0.0144)	-0.00903 (0.0143)
Observations	45,941	45,941	45,941	45,941	45,941	45,941	45,941
Panel B: Dependent variable: perpetrator employment							
Treatment	-0.123 (0.0124)	-0.107 (0.0137)	-0.112 (0.0137)	-0.147 (0.0149)	-0.0886 (0.0104)	-0.128 (0.0123)	-0.147 (0.0149)
Treatment × Female victim	0.0526 (0.0176)	0.0482 (0.0177)	0.0457 (0.0177)	0.0363 (0.0177)			0.0356 (0.0176)
Treatment × Perp is manager			0.105 (0.0168)		0.107 (0.0167)		0.0701 (0.0170)
Treatment × Income gap				0.107 (0.0136)		0.109 (0.0135)	0.0989 (0.0137)
Treatment × Assault		-0.0347 (0.0145)	-0.0429 (0.0146)	-0.0408 (0.0142)	-0.0462 (0.0145)	-0.0434 (0.0141)	-0.0458 (0.0143)
Observations	46,628	46,628	46,628	46,628	46,628	46,628	46,628
Year fixed effects	✓	✓	✓	✓	✓	✓	✓
Time since crime fixed effects	✓	✓	✓	✓	✓	✓	✓
Individual-incidence-year FE	✓	✓	✓	✓	✓	✓	✓
Age by time since crime	✓	✓	✓	✓	✓	✓	✓

Notes. The table reports estimates of equation (2), examining the degree to which power differentials explain the difference in employment effects for perpetrators with male versus female victims. Data are from police reports linked to FLEED administrative register data. Standard errors are clustered at the individual level.

victim and either the income or managerial gap between victims and perpetrators. We estimate:

$$(2) \quad Y_{ibt} = \delta D_{ibt} + \beta D_{ibt} \times \text{Female Victim}_i + \eta D_{ibt} \times \text{Power}_i + \alpha_{ib} + \gamma_t^g + \omega_j^g + \text{Age}_{ib} \times \beta_j + \epsilon_{ibt},$$

where Y_{ibt} is an employment dummy for the victim (perpetrator) i in base-year sample b at time t , D is a treatment indicator equal to one if an individual is a victim or perpetrator of between-colleague violence after the incident occurs. We again include individual-incident-year fixed effects, α_{ib} , year fixed effects, γ_t , time-since-event fixed effects, ω_j and age controls. We allow the year and time-since-event year fixed effects to vary by gender since women have different earnings profiles over time than men. Power is either a dummy taking value of one if the perpetrator is a manager or is equal to the income rank gap between victim and perpetrator.

Table V gives the results. For victims, the coefficient on the interaction between the treatment and a victim's gender is negative but not significantly so. Consistent with Figure IV, there is no statistically significant gender difference in victim effects in the pooled regression, even before controlling for power asymmetries in the relationship between victim and perpetrators (p -value = .383). However, as shown in Figure IV, there is a significant difference in the employment consequences for male perpetrators who attack female versus male colleagues prior to controlling for power differences, with men who attack women 5.3 percentage points more likely to keep their jobs than men who attack men. However, the magnitude of the interaction between the treatment and whether a victim is female reduces by 32.4% (although it remains significant) once we control for the power imbalance between victim and perpetrator and whether the crime is an assault in column (7). Thus, the smaller employment losses for male perpetrators with female victims in Figure IV are partly (but not fully) due to a composition effect, that is, power imbalances are more common for male-female crimes than for male-male ones, and these imbalances play a key role in mediating the effects of workplace violence. Note that columns (2)–(7) show that perpetrators are more likely to enter unemployment if they assault a colleague, consistent with more severe crimes leading to more severe consequences in general. We also show that results are robust to

including controls for the age gap between perpetrator and victim in [Online Appendix Table D.6](#).

Together the results from this section show that female victims are systematically attacked by higher-status male colleagues compared with male victims. This combination—those in power being more insulated from employment effects and men in power attacking lower-ranked and less powerful female colleagues—seems to play a substantial role in explaining the asymmetry in [Figure IV](#). These findings are crucial in terms of the economic and policy implications of this research. They suggest that policies that support women moving up the occupational hierarchy while holding managers and higher-earning men accountable for the mistreatment of subordinates might be key factors in mitigating the negative effects of these events on women.

III.D. Eliminating Other Possible Explanations

The immediate, large, and discontinuous changes in victim and perpetrator labor market outcomes at the point of a violent incident relative to their own prevalence outcomes and those of their matched controls, the fact that these differences persist for at least five years after the incident, and the lack of pretrends showing deterioration in outcomes ahead of the event (see also [Section II.B](#)) rules out many alternative explanations for our findings.

To rule out additional alternative explanations, in [Online Appendix C](#) we provide details of a series of robustness exercises that can be summarized as follows. First, we employ a fuzzier matching approach and show that pretrends remain flat and the effect sizes are similar. Second, we use a different identification strategy instead of the matched DiD, namely, we use future victims as counterfactuals for current victims (and similarly for perpetrators) and find even larger effects. Third, we implement a placebo check where we estimate effects on victims five years prior to workplace violence. We expect this exercise to show no effects, and that is precisely what we find. Together, these additional checks and placebo tests help reassure us that our main results capture the true consequences of violence between colleagues and are not confounded by alternative explanations. We refer the interested reader to [Online Appendix C](#) for more details.

IV. EFFECTS ON THE FIRM

Does between-colleague violence have effects that extend beyond the victim and perpetrator? We first consider measures of the headcount of the firm and whether the firm remains in business ([Online Appendix Table D.8](#)). We find no overall impact of colleague violence on headcount nor on firm death for male-female or male-male violence.

Even if the overall size of the workforce does not change, the composition of workers at affected firms may change if workers with systematically different characteristics leave or join after a violent incident. For example, following an incident of male-female violence in the firm, other women in the firm may be more likely to leave (especially given the low separation rates for perpetrators). Alternatively, hiring rates could be affected. As the majority of perpetrators are men, this could lead the firm to hire fewer men if men in general, rather than just the perpetrator, are punished. Alternatively, women may be seen as creating disruption for firms following male-female violence, adding a friction to their hiring.³³ Thus, we next consider the effect of workplace violence on the share of women employed by the firm in general and not just the effect on the female (or male) victim and perpetrator.

[Figure VII](#), Panel A demonstrates that after an incident of male-female workplace violence, there is a significant decline in the proportion of women employed in firms where such violence took place relative to their matched control firm. The effect is quantitatively large and persistent: relative to the pre-incident mean, the share of women employed by the firm falls over 2 percentage points by five years after the incident. [Figure VII](#), Panel B demonstrates that this effect is isolated to firms where male-female violence occurs. There is no statistically significant effect of male-male violence on the share of women employed in the firm in the four years postviolence (and equivalently, no significant effect on the share of men employed).

Next we explore whether the decrease in the share of women in the firm is explained by more female employees leaving the firm or by the firm hiring fewer women after the incident.

33. This is a similar hypothesis to that explored in [Sarsons \(2017\)](#), who shows that after a female doctor experiences a patient death, referring doctors are less likely to refer cases to women in general, while no such reaction is apparent for male doctors who experience a patient death.

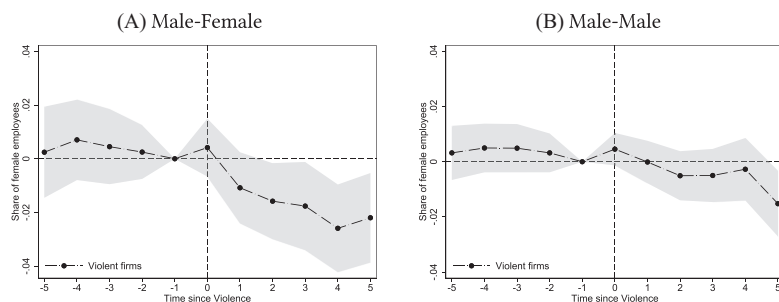


FIGURE VII

Effect on the Share of Female Employees in the Firm

Panel A shows the effect of a violent incident between colleagues on the share of female workers at the firm in which both perpetrator and victim were employed at the time of the incident for male-female between-colleague violence; Panel B shows the same effect but for male-male between-colleague violence. The estimates use the matched-control DiD design to identify effects five years before and five years after a violent incident against a colleague. Panel B shows the effects on the share of female employees in the firm of a “placebo” violent incident where the perpetrator is employed by the firm but the victim is not. The horizontal axis displays time in years. Dashed vertical lines indicate the year of between-colleague violence. Standard errors are clustered at the firm level.

Figure VIII shows that both dimensions are relevant.³⁴ The proportion of women among firm leavers after male-female violence increases (p -value = .148), while the share of women among new hires falls (p -value = .003).³⁵ While our research design removes any time-invariant differences across firms, including time-invariant hiring practices, this is interesting given that firms where male-female violence takes place did not avoid hiring women before the incident (Table III). Instead, these firms had the same share of women employees and were equally likely to hire women compared with all other nonviolent firms prior to the incident.

It is interesting to explore whether the decline in the female share is driven by the presence of a perpetrator of violent crimes in general, who might exhibit a toxic workplace presence,

34. See [Online Appendix Figure E.17](#) for the equivalent results for male-male crimes. As expected the figure shows no significant effects on any dimensions for male-male violence.

35. New hires are workers who we observe in the plant in the December of that year who were not employed by the plant in December of the preceding year.

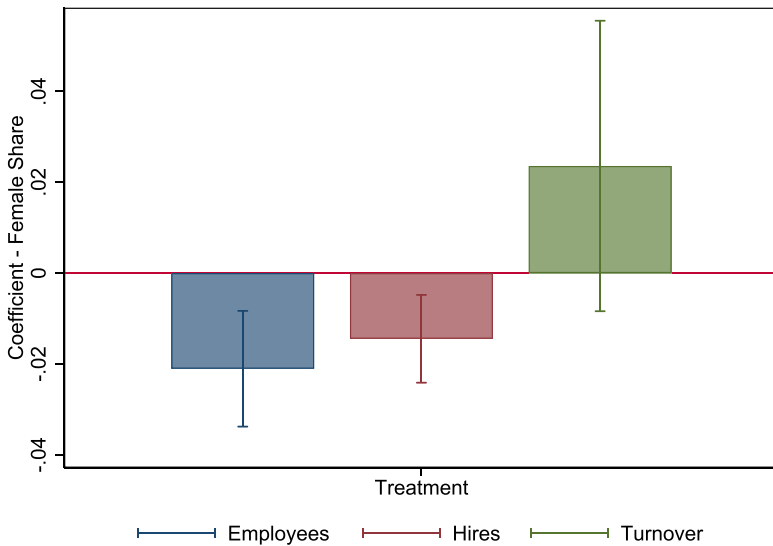


FIGURE VIII

Individual Components of the Drop in the Share of Female Employees for Male-Female Violence

The figure reports DiD estimates of the effect of between-colleague violence on the overall share of women in the firm (in blue, leftmost bar), the share of women among new hires (in red, middle bar), and female turnover in the firm (in green, rightmost bar). Effects are shown for male-female between-colleague violence. Turnover is measured as the share of women among workers leaving the firm. Standard errors are clustered at the firm level.

or whether these spillovers on the broader workforce are linked to the workplace nature of between-colleague violence itself. We explore this in a placebo test in [Figure IX](#), where we estimate the impact of a violent crime where the perpetrator is employed by the firm, but the victim is not, on the share of female employees in the firm. We find precise zeros for both male-female and male-male violence, with no effect on the broader firm from such cases. Thus, our results appear to be uniquely driven by the fact that one colleague attacked another colleague. Because we find no spillovers of noncolleague violence in the broader workforce, for the remainder of the article, which focuses on understanding these spillovers, we do not consider the alternative counterfactual of noncolleague violence.

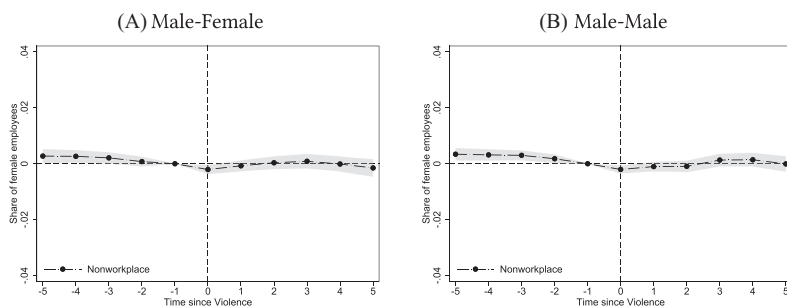


FIGURE IX

Effect of Noncolleague Violence on the Share of Female Employees in the Firm

Panel A shows the effect of a “placebo” violent incident, where the perpetrator is employed by the firm but the victim is not, on the share of female employees in the firm where the perpetrator is employed for male-female between-colleague violence. Panel B shows the same effect but for male-male between-colleague violence. The estimates use the matched-control DiD design to identify effects five years before and five years after a violent incident against a colleague. The horizontal axis displays time in years. Dashed vertical lines indicate the year of between-colleague violence. Standard errors are clustered at the firm level.

IV.A. *Interpreting the Decline in Women Hires*

Falls in the proportion of women hired by the firm could be driven by supply-side factors, that is, women being less likely to apply for jobs at the firm, or by demand-side factors, such as the firm being less likely to hire women from a given set of applicants. We cannot observe the characteristics of applicants to the firm to analyze whether fewer women apply for positions after male-female violence. Instead, we investigate hiring from within the networks of existing employees to capture the potential influence of supply-side factors, albeit imperfectly. Existing employee networks are an obvious way in which potential future hires could find out about and avoid workplaces where one employee attacked another through the “whisper network.”³⁶ A large literature shows that within-network hiring provides an important and high-quality pool of potential applicants to the firm (Marmaros and Sacerdote 2002; Bayer, Ross, and Topa 2008;

36. The supply-side response provides an assessment of the effectiveness of the “whisper network,” a term popularized following #MeToo to define the informal dissemination of information among women. The whisper network is meant to provide information to other women about bad firms and bad actors so that women can avoid them.

Beaman and Magruder 2012; Brown, Setren, and Topa 2016). For example, Barwick et al. (2019) find that 38% (30%) of workers in China (United States) find jobs through personal connections. Moreover, Hensvik and Skans (2016) show that firms are able to hire workers who are of higher quality through referrals from existing highly productive workers.

To test whether within-network hiring decreases after a violent incident, we follow Hensvik and Skans (2016) and construct the set of colleagues that each employee of the firm worked with in some previous employment relationship at some other firm in the 10 years before the incident year. We restrict to plants with 500 workers or fewer to make it more likely that employees in coworking networks actually know each other. We report event study estimates of the effect of between-colleague violence on the share of new hires that come from this network relative to a violent firm's matched control observation in Online Appendix Figure E.18. We find a precise zero impact of a violent incident between colleagues on the amount of within-network hiring done by the violent firm. However, whisper networks about harassers are generally assumed to function primarily between women. Thus, in Online Appendix Figure E.18, Panel B, we reestimate the main results but instead only look at the networks of women employees. We still find no effect on the hiring of future employees from existing women employees' networks.

Although these results suggest that the fall in the female share is not purely driven by supply-side factors, they are not conclusive. The share of new hires who are past colleagues of current employees is quite low (Online Appendix Figure E.19), limiting the power of our test. Further, we cannot observe friend groups or other informal networks that may be even more effective channels for whisper networks. Last, some cases may become public when they reach courts, mitigating the importance of whisper networks, something we cannot observe in police data. Thus, more evidence is necessary to fully understand whether changes in the behavior of applicants or firms drive the fall in female hires following an incident of between-colleague violence.

V. THE IMPORTANCE OF MANAGERS

Firm differences in the management of colleague violence could mediate or accentuate the effect of violence on the wider workforce. Managers play an important role in determining

the success of a firm (Bertrand and Schoar 2003; Bloom and Van Reenen 2007; Bloom et al. 2013; Bandiera et al. 2020). Estimates suggest that differences in management account for 20% of the variation in productivity across plants (Bloom et al. 2019). A smaller literature documents important heterogeneity in how male and female decision makers interpret and respond to negative (or positive) shocks, and how the gender of the individual responsible for the shock might change the response of the manager (Sarsons 2017; Benson, Li, and Shue 2021; Chakraborty and Serra 2021).³⁷ Most closely related to this article, Egan, Matvos, and Seru (2022) find that following incidents of financial misconduct by financial advisers, women who commit such misconduct are more likely to be fired. They find that this disparity in the consequences for financial misconduct by gender are driven entirely by male-managed firms.

Motivated by these facts, we consider heterogeneity in the effect of male-female violence on the share of female workers in the firm by the proportion of women in decision-making positions in the firm. Following Bender et al. (2018), we identify workers in the top 20% of earners in the firm as those with decision-making power. If the proportion of women in the top 20% is above the median, then we label the firm female-managed. On average, women make up 27% of the top 20% of earners in firms where male-female violence occurs, with a standard deviation of 35%. Given that there are significant cross-industry differences in the proportion of women among the highest earners, we control for the industry share of women in the top 20% of all earners (and its interaction with the treatment variable) in all specifications to ensure we capture firm, as opposed to industry, heterogeneity.

Figure X shows that although there is a significant decline in the share of female employees after an incident of male-female violence in male-managed firms relative to their matched control, we see no significant effect on the share of female employees for female-managed firms. The persistence in the fall of women for male-managed firms is particularly striking, with significant negative effects lasting at least five years, compared to no significant effects and estimates close to zero for all years after the incident for female-managed firms. The overall effects are also large, with an almost 6 percentage point decline in the share of

37. See also the literature on how female leadership impacts policy (Ford and Pande 2011; Bertrand et al. 2019).

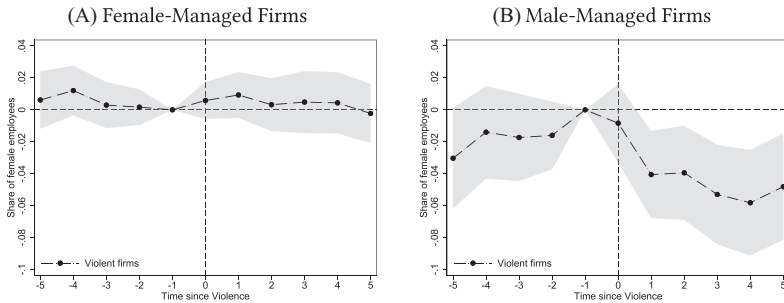


FIGURE X

The Effect on the Share of Female Employees by Firm Management for Male-Female Crimes

The figure shows the effect of male-female violence on the share of female workers in the firm separately for female-managed firms (Panel A) versus male-managed firms (Panel B). We define management as “male” if the share of men in the top 20% of earners is above the median share, and “female” otherwise. Standard errors are clustered at the firm level.

women employed in male-managed firms by five years after male-female violence. This effect is quantitatively significant relative to the baseline share of women employed in these firms of 24.1%. In [Table VII](#) we estimate the role of male versus female management in a stacked DiD regression framework. Column (1) interacts the treatment variable of workplace violence with a dummy variable indicating whether the firm is female-managed. We find that female management is associated with significantly more women in the firm following male-female workplace violence (p -value = .001), such that the gender composition of the firm remains unchanged.

V.A. Differences in Female-Managed Firms

How does female management mediate the effect of male-female violence on the gender composition of the firm? A surprising finding of [Section III](#) was that perpetrators of male-female violence face relatively limited labor market costs compared with perpetrators of male-male colleague violence and perpetrators of male-female noncolleague violence. We thus explore whether perpetrators are more likely to lose their job under female management and how this influences the wider repercussions of violence on the workforce.

TABLE VI
FEMALE MANAGEMENT AND EFFECTS ON VICTIM AND PERPETRATOR
EMPLOYMENT

	Dependent variable: Employment			
	Male-female		Male-male	
	Victim (1)	Perpetrator (2)	Victim (3)	Perpetrator (4)
Treatment × Above-median female management	0.005 (0.019)	−0.055 (0.019)	0.000 (0.020)	−0.084 (0.018)
Treatment	−0.093 (0.015)	−0.043 (0.014)	−0.075 (0.015)	−0.082 (0.014)
Year fixed effects	✓	✓	✓	✓
Time since crime fixed effects	✓	✓	✓	✓
Firm fixed effects	✓	✓	✓	✓
Observations	22,861	23,253	23,076	23,374
Nonviolent mean	0.854	0.867	0.857	0.861

Notes. The table reports the effect of a violent incident between colleagues interacted with whether the firm is female-managed on victim and perpetrator employment outcomes separately for male-female and male-male crimes as specified by the column headings. We define management as “male” if the share of men in the top 20% of earners is above the median share, and “female” otherwise. The reported DiD estimates use the matched control for the victim (or perpetrator) to identify effects five years after versus five years before a violent incident using the event study design, estimating equation (1), but adding the interaction term with female managers. Employment is measured at the end of the year. Standard errors are clustered at the individual level.

We analyze the relationship between female management and the individual labor market effects of violence on victims and perpetrators. Table VI gives the stacked DiD coefficients on the treatment variable and its interaction with a dummy variable for whether the firm is female managed, where the counterfactual is given by individuals’ matched control observations. Victim outcomes are not significantly influenced by the gender composition of management for male and female victims. However, perpetrators have significantly lower employment rates after a violent incident in female-managed firms: for male-female (male-male) violence, perpetrators in female-managed firms have an approximately 5.5 (8.4) percentage point greater reduction in employment compared with their matched control and relative to male-managed firms.³⁸

38. We also find that perpetrators are 4.5 (1.5) percentage points more likely to become not employed under female management following male-male

Thus, female-managed firms appear to be less accommodating of perpetrators of workplace violence in general, and not just for male-female violence. The major difference is that perpetrators of male-female violence in male-managed firms are significantly less likely to fall into unemployment compared with perpetrators of male-male violence. The more negative consequences for perpetrators in female-managed firms could be due to female managers being more likely to fire perpetrators. Another possible explanation is that firms with higher shares of women in their managerial force have relational atmospheres in which it is not tolerable for one colleague to assault another, with perpetrators more likely to choose to leave after an event as a result (e.g., because their relationships with their coworkers deteriorate, their relationships with their managers deteriorate, and/or their promotion possibilities deteriorate).

In Table VII, we jointly examine the effect of perpetrators losing their job and of female management on the share of women employed in the firm after male-female violence. While we showed that perpetrators are more likely to separate from firms with higher shares of female managers following a violent event, which means that the results in this table should be interpreted as suggestive, we still view this exercise as informative. To capture perpetrator job loss, we introduce a dummy variable equal to one if the perpetrator loses their job in the postviolence period. Column (2) demonstrates that following male-female violence, perpetrators losing their job reduces the effect of colleague violence on the share of women employed at the firm (p -value = .006), although this does not fully negate the full effect of the violence. Interestingly, column (3) show that although female-managed firms where the perpetrator remains employed still have different outcomes to male-managed firms (p -value = .081), the perpetrator entering unemployment in either male- or female-managed firms seems to mitigate negative effects on the broader workforce.

1. *Robustness of Management Results.* Female-managed firms may be different on a host of characteristics beyond the gender composition of management. First, we investigate other possible correlates with our measure for female management and

(male-female) between-colleague violence if we use a dummy for whether the manager of the plant is female as opposed to the definition of management based on top earners used in our main specification.

TABLE VII
GENDER COMPOSITION OF MANAGEMENT EFFECT ON SHARE FEMALE EMPLOYEES

	Dependent variable: share female employees in firm					
	Male-female			Male-female with controls		
	(1)	(2)	(3)	(4)	(5)	(6)
Treat × Female management	0.0325 (0.0090)		0.0252 (0.0144)	0.0307 (0.0102)		0.0227 (0.0149)
Treat × Perpetrator job loss		0.0250 (0.0090)	0.0247 (0.0155)		0.0272 (0.0085)	0.0244 (0.0151)
Treat × Female Management × Perp job loss			0.0099 (0.0174)			0.0102 (0.0172)
Treatment	−0.0227 (0.0083)	−0.0387 (0.0085)	−0.0386 (0.0135)	−0.0136 (0.0111)	−0.0393 (0.0113)	−0.0353 (0.0152)
Additional controls				✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Time since crime fixed effects	✓	✓	✓	✓	✓	✓
Firm fixed effects	✓	✓	✓	✓	✓	✓
Observations	14,634	14,634	14,634	13,929	13,929	13,929
Dependent variable mean	0.4042	0.4042	0.4042	0.2083	0.2083	0.2083

Notes. The table reports the effect of a violent incident between colleagues on the share of women employed in the firm. We define management as “male” if the share of men in the top 20% of earners is above the median share, and “female” otherwise. Perpetrator job loss is a dummy variable equal to one if the perpetrator becomes unemployed in the five years after the incident. The reported DiD estimates use the matched control for the firm to identify effects five years after versus five years before a violent incident using the event study design, estimating equation (1), but adding the interaction term with female managers. Standard errors are clustered at the firm level. Additional controls consist of dummies for above- or below-median firm size, average earnings, average age, average tenure, and share female in the firm, where we interact each separately with the treatment.

control for them in the main results. Specifically, we interact the treatment of between-colleague violence with dummy variables indicating above or below median firm size, median earnings, median age, median tenure, and median share female in the firm. Our results are virtually unchanged after including these controls. It is still the case that the perpetrator is much more likely to enter unemployment following violence in female-managed firms ([Online Appendix Table D.9](#), columns (1), (2), (5), and (6)). We also find that our results on the effect of female management on the share female in the firm are virtually unchanged ([Table VII](#), columns (4)–(6)).

In addition, we showed that perpetrators of male-female violence are more likely to be managers themselves. One explanation for our results is that male managers may also be perpetrators and choose not to fire themselves. To explore if this is the operative channel for the female management effects on perpetrator employment, we replicate [Table VI](#) but exclude cases where the perpetrator was also a manager. We report results in [Online Appendix Table D.9](#), columns (3) and (4) for male-female cases and (7) and (8) for male-male cases. We find that even in these cases, female management is associated with a higher likelihood that perpetrators enter unemployment, suggesting that this is not the only channel for these results.

VI. CONCLUSION

We estimated the effects of realized violence between colleagues on victims, perpetrators, and the broader firm. We find that one colleague assaulting another has large negative economic effects on victims and perpetrators. However, male perpetrators of male-female violence experience less severe repercussions compared with perpetrators of male-male violence. Our results show that relative economic power in the firm plays a key role in insulating male perpetrators of violence against women at work. Male-female between-colleague violence has broader implications for women in the firm in general and not just for the female victim. Following male-female violence, firms become significantly more male, with no such repercussions following male-male violence. This is explained by a reduction in the share of female new hires as well as women leaving the firm.

Our results have a number of implications. First, female victims of workplace violence have few economic incentives to report

violence at work. A major issue in preventing harassment at work is that victims rarely report the problem (Magley 2002).³⁹ Understanding who reports, and how reporting can be increased, is a key area for future research (Boudreau et al. 2023). Women underreporting harassment and violence at the hands of a colleague (and in particular one's manager) is easily reconciled with the comparative lack of career consequences for perpetrators of male-female violence we have documented.

Second, given that underreporting is common, we are likely only observing a small fraction of all cases of between-colleague violence, given just 10% of physical assaults are reported to the police in Finland, with lower reporting rates for crimes considered less serious by the victim (European Institute for Crime Prevention and Control 2009; EU Agency for Fundamental Rights 2015). Conservatively, this implies that the incidence of workplace violence is at least 10 times larger than can be documented by police reports. At the same time, underreporting and selective reporting is relevant for the external validity of our results. We might not expect to see quite as large labor market effects on victims, perpetrators, and the firm from less severe abuse by colleagues. However, our effect sizes are equivalent in size to other important economic shocks that have motivated large literatures. For example, in Finland, an exogenous job loss reduces employment over the next six years by 10.9 percentage points (Kaila, Nix, and Riukula 2022). This is only slightly larger than the employment effect for women of being victimized by a colleague from work of 8.4 percentage points. Thus, even if less severe forms of harassment result in only a small fraction of the costs to victims, given that survey evidence suggests somewhere between one-tenth to half of all women experience harassment at work (Folke and Rickne 2022) the overall effects on female employment, female earnings, and the economy as a whole could be very large.

Third, the firm responses we have documented have potentially broader implications for sorting across firms. The fact that male-female violence leads male-managed firms to change their workforce composition toward male employees could partially segment the workforce, leading to male-dominated workplaces

39. "Based on anonymous survey responses, no fewer than 1 in 28 U.S. workers report having been victimized by workplace sexual harassment annually. Yet only 1 in 11,000 workers file a formal sexual harassment charge with the Equal Employment Opportunity Commission (EEOC), the agency tasked with enforcing all federal antidiscrimination laws" (Dahl and Knepper 2021, 1).

where male management repeatedly allows perpetrators of male-female violence to remain employed at the expense of female employees. This would likely entail an equilibrium in which women in the firm's gender minority are also more likely to find themselves in firms that tolerate abuse and harassment of women, consistent with the descriptive facts documented in [Folke and Rickne \(2022\)](#).

Last, our results lead to the question of what can be done to mitigate the costs of these events. The results from this article provide one optimistic takeaway: female-managed firms are able to mitigate the effects of male-female workplace violence on other female employees in the firm. They accomplish this in part by being more likely to force the perpetrators of these crimes into unemployment. One possible interpretation is that female managers have less tolerance for misbehavior, regardless of the gender of the misbehaving party. Alternatively, these differences could be consistent with an "in-group" tolerance. In other words, male managers may be more forgiving of male perpetrators. This latter explanation would be consistent with the findings from [Egan, Matvos, and Seru \(2022\)](#) and [Cullen and Perez-Truglia \(2019\)](#).

In any case, these results suggest that there is a way to reduce the costs of violence against women at work, namely, by ensuring that violent actions against colleagues result in consequences for the perpetrators. Our results demonstrate that this is not done as frequently among male-managed firms. Such lack of consequences not only benefits perpetrators at the cost of their victims but is also costly to women in general, as they are less likely to be employed in these relatively high-paying firms in the future. However, our analysis cannot predict if firing perpetrators would increase the profits of these firms. If the perpetrators are highly productive workers, then there may be high economic benefits to retaining them, presenting a difficult decision for managers. Such decisions would likely also depend on a notion of fairness and justice, which is well beyond the scope of our analysis.

Our results suggest several avenues for new research. First, data constraints make it impossible for us to explore the effects of lower-level bullying and harassment because they are not reported to the police. However, obtaining such data and understanding if lower-level harassment has similar effects on perpetrators, victims, and the broader firm would be informative. Second, our results add to the growing literature on the importance of who is in management. Understanding differences

in how male versus female decision makers manage firms beyond just conflict between colleagues could reveal other important implications for the workforce. Last, our analysis suggests that there is a “business case” for preventing violence and harassment against women, beyond the obvious ethical one, although we do not quantify this cost. We show that turnover rates of women and hiring are significantly affected by male-female violence in male-managed firms. In the face of turnover and hiring costs, these changes could potentially be very costly, beyond the effect on the loss of talent.

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SUPPLEMENTARY MATERIAL

An [Online Appendix](#) for this article can be found at *The Quarterly Journal of Economics* online.

DATA AVAILABILITY

The code underlying this article is available in the Harvard Dataverse, <https://doi.org/10.7910/DVN/ENNRDG> (Adams-Prassl et al. 2023).

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